NOAA Technical Memorandum NMFS-XXX-##

Analyses to support the Dolphinfish Management Strategy Evaluation in the U.S. Atlantic

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January 2023

U.S. DEPARTMENT OF COMMERCE

National Oceanic and Atmospheric Administration
National Marine Fisheries Service
Northwest Fisheries Science Center

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1 Introduction

Here we will be displaying various data summaries and analyses related to dolphin management in the U.S. South Atlantic.

1.1 Details

The documentation of this data exploration was developed in R, using R Studio and Quarto.

2 Defining the regions for the operating model

Here we define the different regions to be used in the operating model, based on the areas that were defined in the industry working groups. The operating model is intended to encompass the Western Central Atlantic dolphin population and include international waters (FAO areas 21 and 31) which are thought to be connected with the stock that is exploited by U.S. fisheries.

We use the U.S. exclusive economic zone (EEZ) to differentiate between national and international waters. Note that some U.S. commercial activity occurs in high seas waters and we will account for those catches in the summaries.

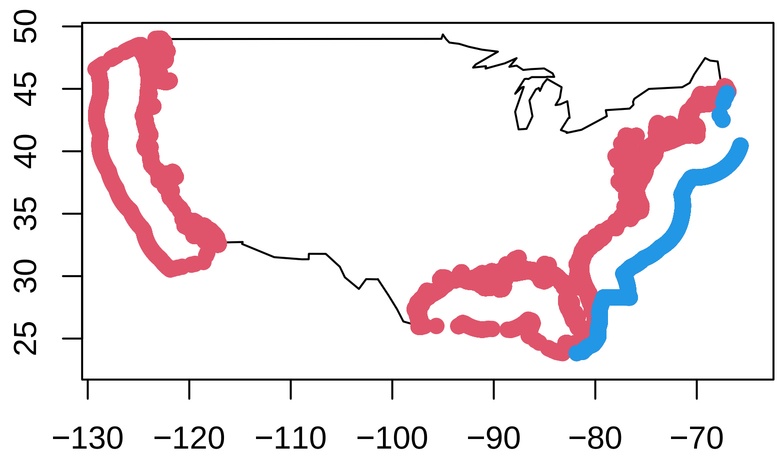
2.1 Data download for the U.S. EEZ

We download the U.S. Atlantic EEZ from Marineregions.org.

```
# clear workspace and load the libraries
if (!requireNamespace("maps", quietly = TRUE)) {
  install.packages("maps") }
library(maps)

rm(list = ls())

# read in the shapefile downloaded from Marineregions.org
#library(sf)
#shp_data <- st_read("data/eez/eez.shp")
#coords <- st_coordinates(shp_data)
#co <- as.data.frame(coords)
#save(co, file = "data/eez/Marineregions_EEZ.RData")

load("data/eez/Marineregions_EEZ.RData")
head(co)
```

```
          X        Y L1 L2 L3
1 -67.28403 45.19125  1  1  1
2 -67.28400 45.19126  1  1  1
3 -67.28387 45.19125  1  1  1
4 -67.28383 45.19125  1  1  1
5 -67.28372 45.19125  1  1  1
6 -67.28365 45.19125  1  1  1
```

```
map("usa", xlim = c(-130, -63), ylim = c(22, 50))
axis(1); axis(2); box()
points(co$X, co$Y, pch = 19, col = 2)

# remove points that are not in the U.S. South Atlantic
co <- co[which(co$X > (-82)), ]
co <- co[-which(co$X < (-81) & co$Y > 25 & co$Y < 28), ]

#plot(co$X, co$Y, col = 0)
#text(co$X, co$Y, 1:nrow(co), cex = 0.7)
#plot(co$X, co$Y, col = 0, xlim = c(-85, -80), ylim = c(22, 25))
#text(co$X, co$Y, 1:nrow(co), cex = 0.7)

# identify the northernmost and southernmost points along the Atlantic EEZ
co1 <- co[2100:3727, ]
points(co1$X, co1$Y, col = 4, pch = 19)
```

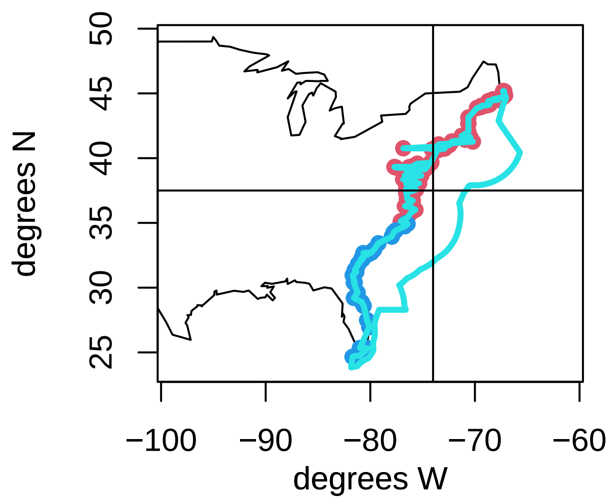


2.2 Construct the U.S. EEZ

We simplify the shapefile along the U.S. coast in order to simplify computations later in the process.

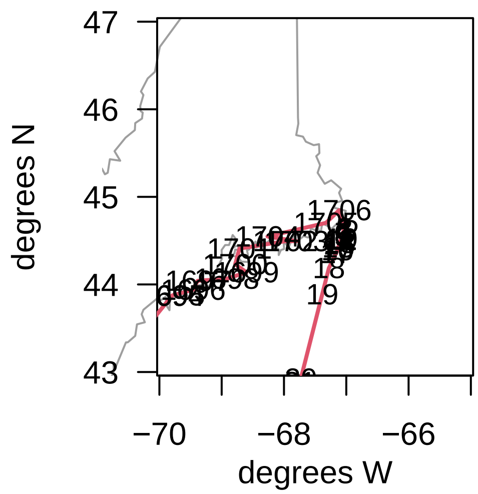
```
# subset points N and S of 35N for later delineation
co2 <- co[-c(2100:3727), ]
co3 <- co2[which(co2$Y > 35), ]
co4 <- co2[which(co2$Y <= 35), ]

# reduce number of points along the coast to make computation easier
co3 <- co3[seq(1, nrow(co3), 1100), ]
co4 <- co4[seq(1, nrow(co4), 800), ]

map("usa", xlim = c(-100, -60), ylim = c(23, 50))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2); box()
points(co3$X, co3$Y, col = 2, pch = 19)
```

```
points(co4$X, co4$Y, col = 4, pch = 19)

co2 <- rbind(co3, co4)
co2 <- co2[order(co2$Y), ]
cofin <- rbind(co1, co2)

polygon(cofin$X, cofin$Y, border = 5, lwd = 3)

# make the polygon along the coast line a bit more linear
abline(h = 37.5)
abline(v = -74)
```

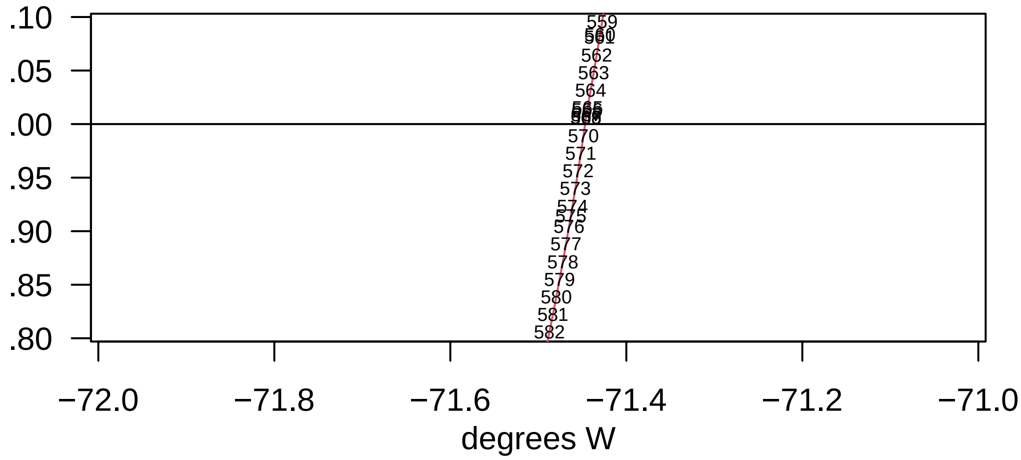



```
cofin <- cofin[-which(cofin$X < (-74) & cofin$Y > 37.6), ]
#text(cofin$X, cofin$Y, 1:nrow(cofin), cex = 0.9)
lis <- c(1634, 1639, 1688, 1710, 1711, 1712)
cofin <- cofin[-lis, ]

map(database = "usa", xlim = c(-70, -65), ylim = c(43, 47), col = 8)
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
```

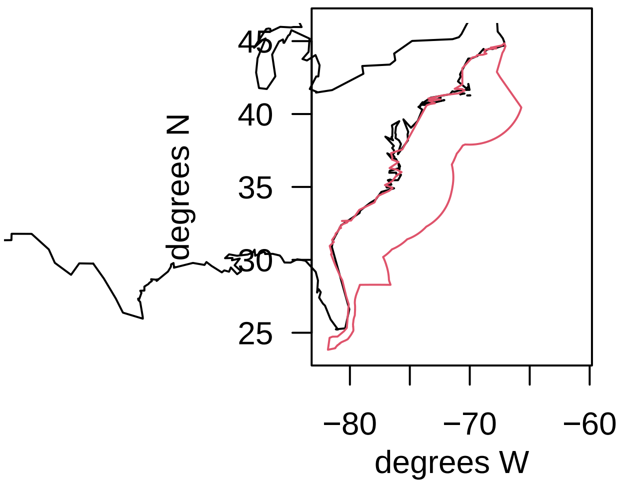
```
polygon(cofin$X, cofin$Y, border = 2, lwd = 2)
text(cofin$X, cofin$Y, 1:nrow(cofin), cex = 0.9)
```




```
map(database = "usa", xlim = c(-72, -71), ylim = c(34.8, 35.1))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(cofin$X, cofin$Y, border = 2)
text(cofin$X, cofin$Y, 1:nrow(cofin), cex = 0.6)
abline(h=35)
abline(v = -60)
```




```r
cofin <- cofin[, 1:2]

map(database = "usa", xlim = c(-83, -60), ylim = c(23, 47), col = 0)
map(database = "usa", xlim = c(-83, -60), ylim = c(23, 47), add = T)
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(cofin$X, cofin$Y, border = 2)
```




```
# change the southernmost tip to exactly 24N so that it alighs with Caribbean polygon
cofin <- cofin[-which(cofin$Y < 24), ]
cofin[1624, 2] <- 24

USeez <- cofin
#write.csv(USeez, file = "eez.csv")
```

Here is a final U.S. South Atlantic EEZ polygon, with a simplified coastline to reduce computational time when subsetting data points.

2.3 Define the Caribbean region

Now we define the boundaries of the Caribbean region. Its northernmost boundary aligns with the EEZ along Florida's coast up to 28.3 degrees N, at which point the boundary cuts southeast until it intersects with the 24N boundary just east of the Bahamas, approximating the Bahamas EEZ. Its western boundary is -90 degrees W and its southern boundary is 8 degrees N (essentially the land boundary of the Caribbean Sea). Its eastern boundary is -71.1 degrees W.

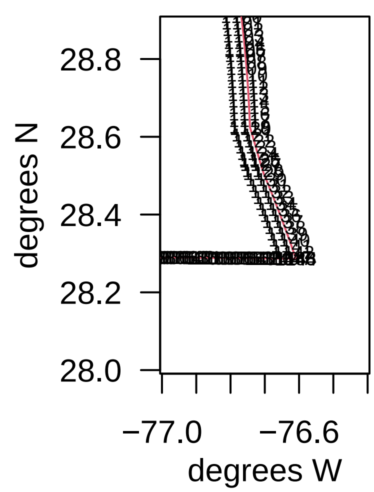
```
which.min(cofin$Y)
```

```
[1] 1624
```

```
cofin[1624, ]
```

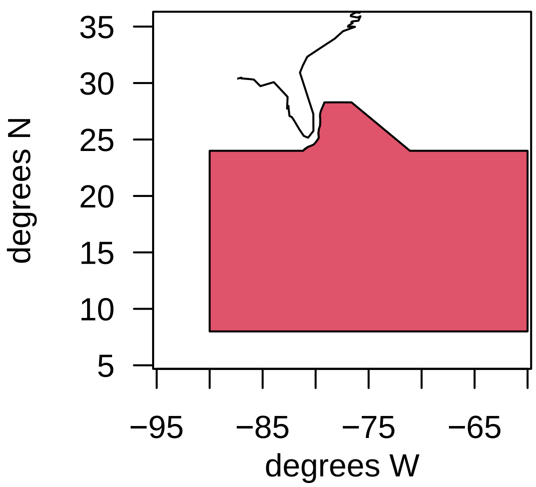
```
             X  Y
3723 -81.16148 24
```

```
map(database = "usa", xlim = c(-77, -76.4), ylim = c(28, 28.9))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(cofin$X, cofin$Y, border = 2)
text(cofin$X, cofin$Y, 1:nrow(cofin), cex = 0.6)
```




```
car1 <- cofin[1143:1624, ]
car2 <- data.frame(cbind(c(-90, -90, -60, -60, -71.1), c(24, 8, 8, 24, 24)))
names(car2) <- c("X", "Y")
CAR <- rbind(car1, car2)
```

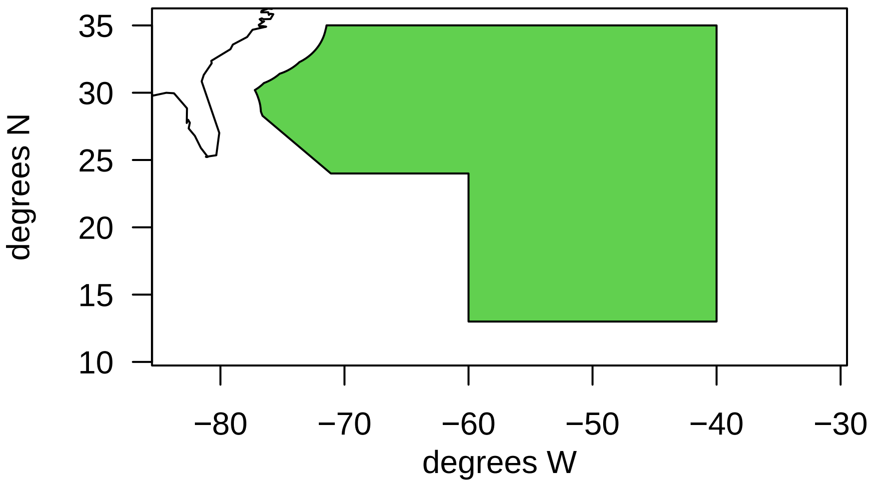
```
names(CAR) <- c("X", "Y")

map(database = "usa", xlim = c(-95, -60), ylim = c(5, 36))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(CAR$X, CAR$Y, col = 2)
```



3 Define the NCA region

The North Central Atlantic (NCA) region, otherwise known as the Western Central Atlantic, is defined according to the boundaries of FAO area 31, but excluding the U.S. EEZ and Caribbean territorial seas to the West.

```
nca1 <- cofin[569:1143, ]  # subset points along EEZ
nca2 <- data.frame(cbind(c(-71.1, -60, -60, -40, -40), c(24, 24, 13, 13, 35))) # define bou
names(nca2) <- c("X", "Y")
NCA <- rbind(nca1, nca2)
names(NCA) <- c("X", "Y")

map(database = "usa", xlim = c(-85, -30), ylim = c(10, 36))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(NCA$X, NCA$Y, col = 3)
```


degrees N
35
30
25
20
15
10
−80
−70
−60
−50
−40
−30
degrees W

4 Define the NED region

The Northeast Distant Waters (NED) region, otherwise known as the Northwesetern Atlantic, is defined according to the boundaries of FAO area 21, but excluding the U.S. EEZ territorial seas to the West. It does include Canadian and French territorial EEZs. The northern boundary is 60 degrees north, as dolphin are not caught above this latitude.

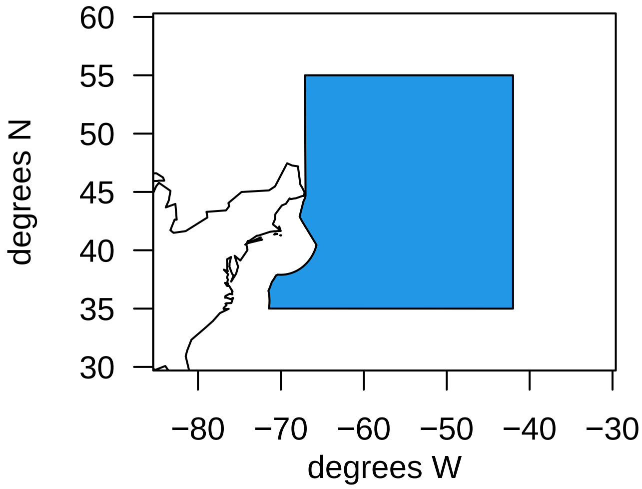
```
which.max(cofin$Y)  # find northernmost point of U.S. EEZ
```

```
[1] 1702
```

```
cofin[1702, ]
```

```
              X        Y
77844 -67.11553 44.84764
```

```
ned1 <- cofin[1:569, ]  # subset points along EEZ
ned2 <- data.frame(cbind(c(-42, -42, -67.1), c(35, 55, 55)))  # define boundaries according
names(ned2) <- c("X", "Y")
NED <- rbind(ned1, ned2)
names(NED) <- c("X", "Y")

map(database = "usa", xlim = c(-85, -30), ylim = c(30, 60))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(NED$X, NED$Y, col = 4)
```


60
55
50
45
40
35
30
degrees N
−80
−70
−60
−50
−40
−30
degrees W

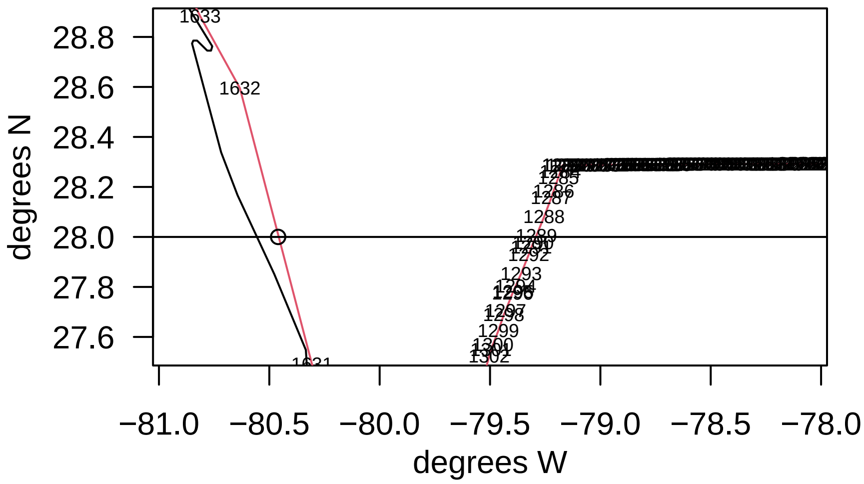
5 Define the Florida Keys region

The southernmost region of the U.S. EEZ is called the FLK region, and includes waters within the EEZ boundary, north up to 28 degrees N. The boundary of 28N is chosen because it aligns with commercial logbook reporting boundaries and aligns roughly with the Brevard - Indian River County border. This is convenient because U.S. recreational statistics are summarized as groups of counties, with SE FL encompassing Miami-Dade to Indian River County and NE FL encompassing Brevard to Nassau County.

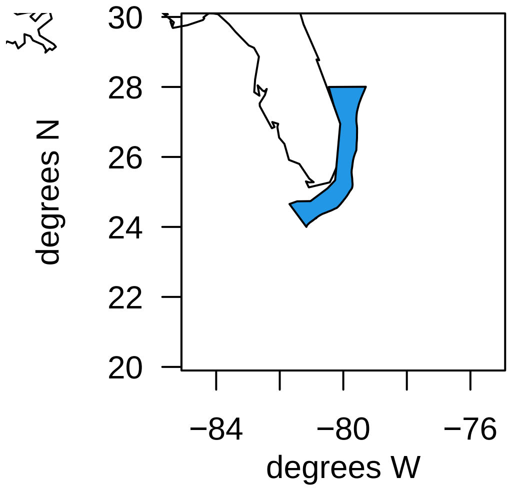
```
which.min(abs(co1$Y - 28))  # find the southernmost point
```

```
[1] 1289
```

```
map(database = "usa", xlim = c(-81, -78), ylim = c(27.5, 28.9))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(cofin$X, cofin$Y, border = 2)
text(cofin$X, cofin$Y, 1:nrow(cofin), cex = 0.6)   # find points falling at 28N
points(-80.46, 28)
cofin[1632, ] <- c(-80.46, 28)
abline(h = 28)
```



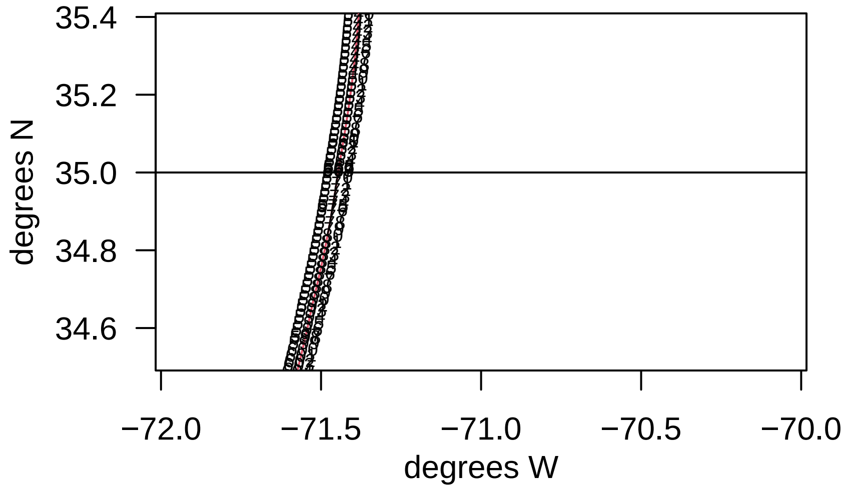

```
FLK <- cofin[1289:1632, ]  # subset points along EEZ

map(database = "usa", xlim = c(-85, -75), ylim = c(20, 30))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(FLK$X, FLK$Y, col = 4)
```


30
28
26
24
22
20
degrees N
−84
−80
−76
degrees W

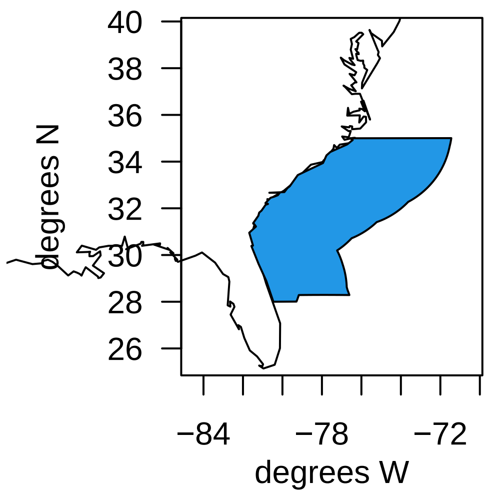
6 Define the North Florida - South NC region

The second area that was decided upon to be defined within the operating model was northern Florida to Southern North Carolina. We form a boundary at 35 degrees north because this is the delineation between the NED and NCA regions. Also, 35N is just south of Cape Hatteras, which is a boundary for NMFS recreational data collection, as well as a boundary of commercial logbook data reporting. Therefore, all sources of data (international, U.S. commercial, and U.S. recreational) can be easily parsed along this boundary.

```
map(database = "usa", xlim = c(-72, -70), ylim = c(34.5, 35.4))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(cofin$X, cofin$Y, border = 2)
text(cofin$X, cofin$Y, 1:nrow(cofin), cex = 0.6)  # find points that fall at 35N
abline(h = 35)
```



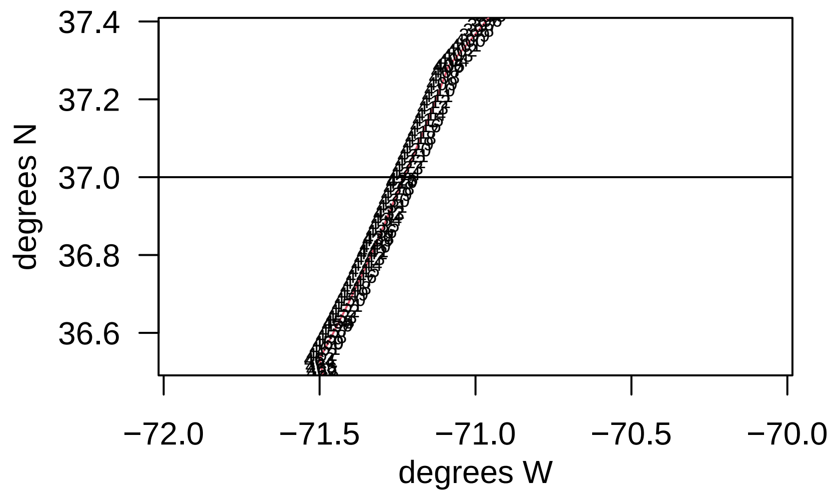

```
cofin[1663, 2] <- 35
NCFL <- rbind(cofin[1632:1663, ], cofin[569:1289, ])  # subset along EEZ

map(database = "usa", xlim = c(-85, -70), ylim = c(25, 40))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(NCFL$X, NCFL$Y, col = 4)
```


40
38
36
34
32
30
28
26
degrees N
−84
−78
−72
degrees W

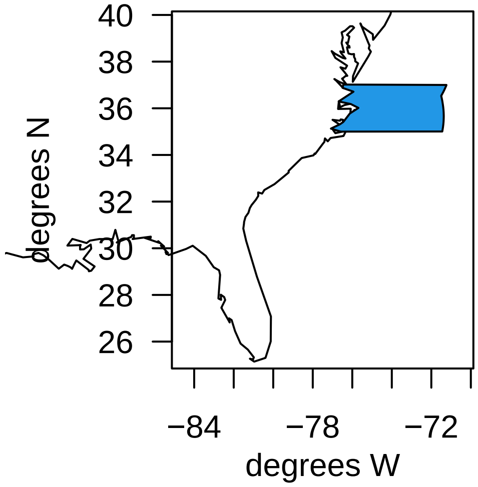
7 Define the northern North Carolina region

The third U.S. area that was decided upon to be defined within the operating model was northern North Carolina, which was intended to encompass the northern part of the state up to the Virginia border. We form a boundary at 37 degrees north because this is a boundary for commercial logbook data reporting and roughly aligns with the latitude of the state boundary between North Carolina and Virginia. Note there is a slight misalignment between this boundary and the way that U.S. recreational statistics are reported (by state), so the commercial and recreational data will be subset along slightly different boundaries for this region.

```
map(database = "usa", xlim = c(-72, -70), ylim = c(36.5, 37.4))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(cofin$X, cofin$Y, border = 2)
text(cofin$X, cofin$Y, 1:nrow(cofin), cex = 0.6)
abline(h = 37)
```



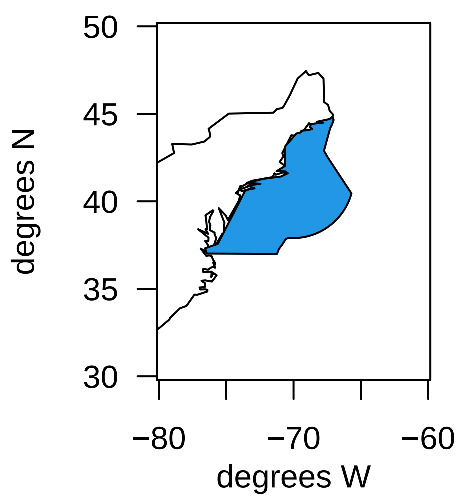

```
cofin[425, 2] <- 37
NNC <- rbind(cofin[1663:1673, ], cofin[425:569, ])

map(database = "usa", xlim = c(-85, -70), ylim = c(25, 40))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(NNC$X, NNC$Y, col = 4)
```


40
38
36
34
32
30
28
26
degrees N
−84
−78
−72
degrees W

8 Define the Virginia and north region

The fourth region within the U.S. EEZ is the remainder of the area, which encompasses national waters from Virginia up to Maine.

```
VBM <- rbind(cofin[1673:1702, ], cofin[1:425, ])

map(database = "usa", xlim = c(-80, -60), ylim = c(30, 50))
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()
polygon(VBM$X, VBM$Y, col = 4)
```



8.1 Plot all of the regions

Now we will put all of the regions on a single map and check that the boundaries align.

```
png(filename = "data/eez/new_spatial_zones1.png", width= 800, height = 900)

map(database = "world", xlim = c(-92, -38), ylim = c(5, 55), col = 8)
mtext(side = 1, "degrees W", line = 2); mtext(side = 2, "degrees N", line = 3)
axis(1); axis(2, las = 2); box()

polygon(CAR$X, CAR$Y, border = 1)
polygon(NCA$X, NCA$Y, border = 1)
polygon(NED$X, NED$Y, border = 1)
polygon(FLK$X, FLK$Y, border = 1)
polygon(NCFL$X, NCFL$Y, border = 1)
polygon(NNC$X, NNC$Y, border = 1)
polygon(VBM$X, VBM$Y, border = 1)

map(database = "world", add = T, col = 8, lwd = 2)

text(-75, 15, "CAR", cex = 1.2)
text(-57, 28, "NCA", cex = 1.2)
text(-57, 42, "NED", cex = 1.2)
text(mean(FLK$X), mean(FLK$Y), "FLK", cex = 1.2)
text(-79.3, mean(NCFL$Y), "NCFL", cex = 1.2)
text(-73.8, mean(NNC$Y), "NNC", cex = 1.2)
text(-71.5, mean(VBM$Y), "VBM", cex = 1.2)

dev.off()
```

```
pdf
  2
```

```
#class(CAR)

CAR$region <- "CAR"
FLK$region <- "FLK"
NCFL$region <- "NCFL"
```

```
NNC$region <- "NNC"
VBM$region <- "VBM"
NCA$region <- "NCA"
NED$region <- "NED"

zones <- rbind(FLK, NCFL, NNC, VBM, NED, NCA, CAR)

# plot(zones$X, zones$Y, col = as.numeric(as.factor(zones$region)))

# output the coordinates to a .csv file
write.csv(zones, file = "data/eez/dolphin_OM_areas.csv", row.names = FALSE)
```

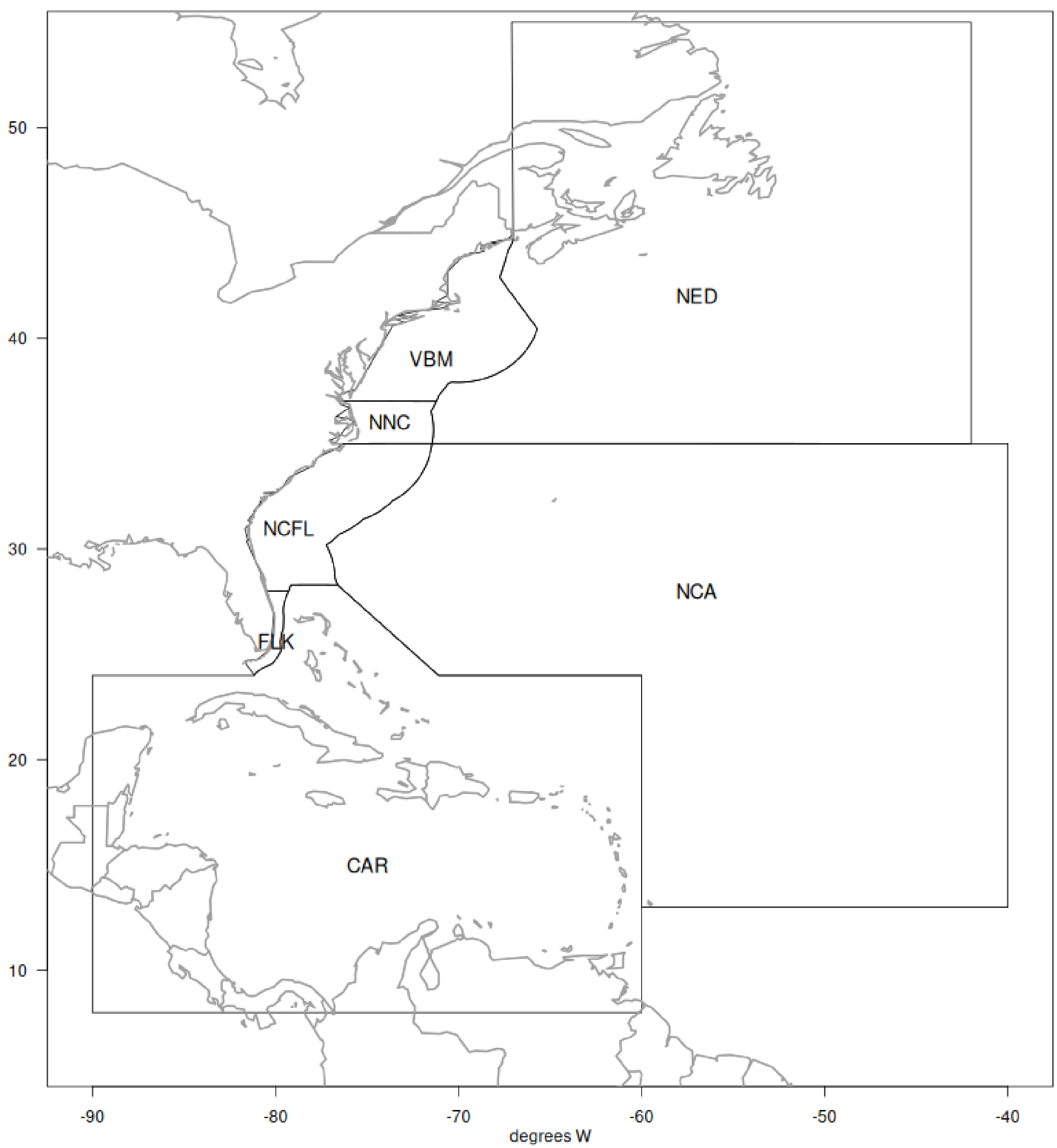


Figure 8.1: Areas defined in the operating model.

9 NOAA Fisheries recreational catch data

Here we will partition recreational catch data from NOAA Fisheries' Marine Recreational Information Program (MRIP), a standardized, peer-reviewed suite of recreational fishing surveys used to produce catch and effort estimates for quota monitoring and stock assessment in the United States. MRIP is a standardized survey suite that employs a complex statistical sampling design stratified across time, regions of the coast, and different fishing modes, which allows the intercept data to be scaled up by effort estimates to calculate total catches.

9.1 Data access and upload

Data are downloaded directly from the Miami server using the files provided to SERO for ACL monitoring. The current location is: M://SFD/SECM-SFD/ACL/2025_Jun23_-MRIP_MRFSS/. The catches are in FES MRIP and the units are in pounds whole weight.

```
# clear workspace
rm(list = ls())

# read in data file
load("data/mrip_fes_rec81_25wv2_23June25.RData")  # most recent file on SFD server
head(dat)
```

```
  REC_ACL_MRIP_FES_ID YEAR MONTH WAVE SUB_REG NEW_ST NEW_STA NEW_MODE NEW_MODEN
1            73407425 1989    NA    4       6      9      NC        4      Priv
2            73407426 1989    NA    4       6      9      NC        4      Priv
3            73407427 1989    NA    4       6      9      NC        4      Priv
4            73407428 1989    NA    4       6      9      NC        4      Priv
5            73407429 1989    NA    4       6      9      NC        4      Priv
6            73407430 1989    NA    4       6      9      NC        4      Priv
  AREA_X NEW_AREA  NEW_AREAN FL_REG NC_REG   DS              NEW_SCI
1      5        5    Inshore     NA      S MRIP Centropristis striata
2      5        5    Inshore     NA      N MRIP Centropristis striata
```

```
3      2        2  Ocean>3mi     NA      N MRIP Centropristis striata
4      2        2  Ocean>3mi     NA      S MRIP Centropristis striata
5      1        1 Ocean<=3mi     NA      N MRIP Centropristis striata
6      1        1 Ocean<=3mi     NA      S MRIP Centropristis striata
         NEW_COM    SP_CODE SPECIES_ITIS SPECIES        AB1          B2
1 black sea bass 8835020301       167687         34181.4033 121019.1368
2 black sea bass 8835020301       167687           795.8612   4445.0527
3 black sea bass 8835020301       167687           549.5390   2919.4840
4 black sea bass 8835020301       167687         71823.2390  24548.2535
5 black sea bass 8835020301       167687           415.7642     87.5928
6 black sea bass 8835020301       167687         48728.5632  11859.6428
           A         B1 LBSEST_SECWWT LBSEST_SECSOURCE SAMPLE_SIZE_USED
1 31761.4755  2419.9278    11457.8425          srysmwa               41
2   584.4155   211.4457      266.7782          srysmwa               41
3   549.5390     0.0000      484.4319          srysmwa              119
4 33343.1844 38480.0546    63313.9276          srysmwa              119
5   415.7642     0.0000      294.0623          srysmwa               71
6 40585.0152  8143.5480    34464.8156          srysmwa               71
   AVG_LEN   AVG_WGT SRYSMWA_AVG_LEN SRYSMWA_AVG_WGT SRYSMWA_SAMPLE_SIZE
1 203.4572 0.3352069        203.4572       0.3352069                  41
2 203.4572 0.3352069        203.4572       0.3352069                  41
3 296.2590 0.8815243        296.2590       0.8815243                 119
4 296.2590 0.8815243        296.2590       0.8815243                 119
5 261.9949 0.7072816        261.9949       0.7072816                  71
6 261.9949 0.7072816        261.9949       0.7072816                  71
  SRYSMWA_FLAGGED SRYSMW_AVG_LEN SRYSMW_AVG_WGT SRYSMW_SAMPLE_SIZE
1              NA       269.2563      0.7310037                231
2              NA       269.2563      0.7310037                231
3              NA       269.2563      0.7310037                231
4              NA       269.2563      0.7310037                231
5              NA       269.2563      0.7310037                231
6              NA       269.2563      0.7310037                231
  SRYSMW_FLAGGED SRYSM_AVG_LEN SRYSM_AVG_WGT SRYSM_SAMPLE_SIZE SRYSM_FLAGGED
1             NA      259.7694     0.6948228               556            NA
2             NA      259.7694     0.6948228               556            NA
3             NA      259.7694     0.6948228               556            NA
4             NA      259.7694     0.6948228               556            NA
5             NA      259.7694     0.6948228               556            NA
6             NA      259.7694     0.6948228               556            NA
  SRYS_AVG_LEN SRYS_AVG_WGT SRYS_SAMPLE_SIZE SRYS_FLAGGED SRY_AVG_LEN
1      270.955    0.8099641              982           NA    269.1276
```

```
2     270.955   0.8099641             982           NA    269.1276
3     270.955   0.8099641             982           NA    269.1276
4     270.955   0.8099641             982           NA    269.1276
5     270.955   0.8099641             982           NA    269.1276
6     270.955   0.8099641             982           NA    269.1276
  SRY_AVG_WGT SRY_SAMPLE_SIZE SRY_FLAGGED SR_AVG_LEN SR_AVG_WGT SR_SAMPLE_SIZE
1   0.8085211            1703          NA   303.6871   1.050403          46055
2   0.8085211            1703          NA   303.6871   1.050403          46055
3   0.8085211            1703          NA   303.6871   1.050403          46055
4   0.8085211            1703          NA   303.6871   1.050403          46055
5   0.8085211            1703          NA   303.6871   1.050403          46055
6   0.8085211            1703          NA   303.6871   1.050403          46055
  SR_FLAGGED S_AVG_LEN S_AVG_WGT S_SAMPLE_SIZE S_FLAGGED     VAR_B2   VAR_AB1
1         NA  344.4373  1.364082        305089        NA 4072492794 310099094
2         NA  344.4373  1.364082        305089        NA    2866289    258947
3         NA  344.4373  1.364082        305089        NA          0         0
4         NA  344.4373  1.364082        305089        NA  105731413 534702898
5         NA  344.4373  1.364082        305089        NA          0         0
6         NA  344.4373  1.364082        305089        NA   22265768 286196169
        SA_LABEL GOM_LABEL REC_ACL MIGRA_GRP AGG_MODEN JURISDICTION PS_REG
1 Snapper Grouper                1             Private        State      6
2 Snapper Grouper                1             Private        State      5
3 Snapper Grouper                1             Private      Federal      5
4 Snapper Grouper                1             Private      Federal      6
5 Snapper Grouper                1             Private        State      5
6 Snapper Grouper                1             Private        State      6
  COUNCILREG GFMC_FMP SAFMC_FMU LBSEST_SECGWT AGGR_AREA AREA FIRST_MONTH
1          6                  Y     9710.0360             NA          NA
2          5                  Y      226.0832             NA          NA
3          5                  Y      410.5355             NA          NA
4          6                  Y    53655.8708             NA          NA
5          5                  Y      249.2054             NA          NA
6          6                  Y    29207.4709             NA          NA
  LAST_MONTH ALT_FLAG CHTS_CL CHTS_H CHTS_RL CHTS_WAB1C CHTS_WAB1H MODE_FX
1         NA        0       0      0       0          0          0       7
2         NA        0       0      0       0          0          0       7
3         NA        0       0      0       0          0          0       7
4         NA        0       0      0       0          0          0       7
5         NA        0       0      0       0          0          0       7
6         NA        0       0      0       0          0          0       7
  NEW_SEASN SEASON
```

```
1                NA
2                NA
3                NA
4                NA
5                NA
6                NA
```

```
#apply(dat[2:18], 2, table, useNA = "always")

# confirm nomenclature for dolphin and subset by species --------------------------
table(dat$NEW_COM[grep("dolphin", dat$NEW_COM)])
```

```

dolphin
   9377
```

```
table(dat$NEW_SCI[grep("dolphin", dat$NEW_COM)])
```

```

Coryphaena hippurus
               9377
```

```
d <- dat[which(dat$NEW_COM == "dolphin"), ]
#apply(d[2:18], 2, table, useNA = "always")
d <- d[which(d$YEAR < 2025), ]    # take out 2025 because incomplete year
```

9.2 Separate Gulf and Atlantic recreational catches

Now we filter the data set to separate out the Gulf versus Atlantic catches.

```
# codes for WFL regions: 1 (NW FL Panhandle- Escambia to Dixie co.), 2 (SW FL Peninsula- Le
# subregional codes: 6 is Atlantic and 7 is Gulf
table(d$FL_REG, d$SUB_REG, useNA = "always")
```

```
         4    5    6    7 <NA>
  1      0    0    0  387    0
  2      0    0    0  110    0
  3      0    0    0  568    0
  4      0    0 1022    0    0
  5      0    0  312    0    0
  <NA>  96  585 3836 2426    0
```

```
unique(d$NEW_STA)
```

```
 [1] "NC"     "FLE"    "FLW"    "SC"     "GA"     "MS"     "AL"     "LA"
 [9] "TX"     "FLE/GA" "RI"     "VA"     "DE"     "MD"     "NJ"     "CT"
[17] "MA"     "NY"     "LA/MS"  "AL/FLW"
```

```
lis <- c("TX", "LA", "LA/MS", "MS", "AL", "AL/FLW")  # filter out the Gulf states
d1 <- d[-which(d$NEW_STA %in% lis), ]
unique(d1$NEW_STA)
```

```
 [1] "NC"     "FLE"    "FLW"    "SC"     "GA"     "FLE/GA" "RI"     "VA"
 [9] "DE"     "MD"     "NJ"     "CT"     "MA"     "NY"
```

```
table(d1$FL_REG, d1$NEW_STA, useNA = "always")
```

```
        CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  1      0    0    0      0  387    0    0    0    0    0    0    0    0    0
  2      0    0    0      0  110    0    0    0    0    0    0    0    0    0
  3      0    0    0      0  568    0    0    0    0    0    0    0    0    0
  4      0    0 1022      0    0    0    0    0    0    0    0    0    0    0
  5      0    0  312      0    0    0    0    0    0    0    0    0    0    0
  <NA>  14  114    0   1794  174   54   20  125 1344  108   73   62  644  165

      <NA>
  1      0
  2      0
  3      0
  4      0
  5      0
  <NA>   0
```

```
dim(d1)
```

```
[1] 7090   83
```

```
d1 <- d1[-which(d1$FL_REG == 1 | d1$FL_REG == 2), ]   # filter out FL regions 1 and 2 but r
dim(d1)
```

```
[1] 6593   83
```

```
d1 <- d1[-which(is.na(d1$FL_REG) & d1$NEW_STA == "FLW"), ]  # filter out FLW with unknown F
dim(d1)
```

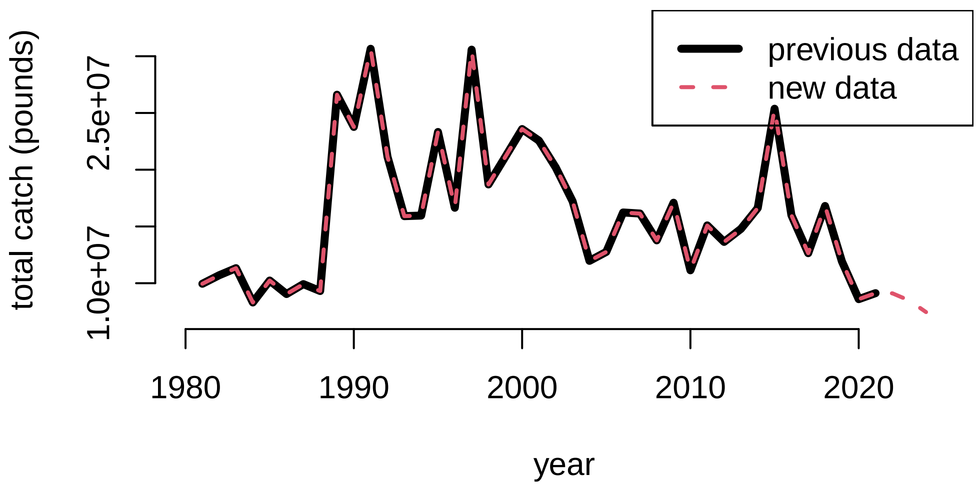
```
[1] 6419   83
```

```
table(d1$FL_REG, d1$NEW_STA, useNA = "always")  # retained are all Atlantic states plus reg
```

```
        CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  3      0    0    0      0  568    0    0    0    0    0    0    0    0    0
  4      0    0 1022      0    0    0    0    0    0    0    0    0    0    0
  5      0    0  312      0    0    0    0    0    0    0    0    0    0    0
  <NA>  14  114    0   1794    0   54   20  125 1344  108   73   62  644  165

       <NA>
  3       0
  4       0
  5       0
  <NA>    0
```

Now we can summarize the total dolphin catches in weight for the U.S. Atlantic, and compare to previous values sent for quota monitoring. We see that the values are almost identical.

```
# summarize by year and compare --------------------------

tot1 <- tapply(d1$LBSEST_SECWWT, d1$YEAR, sum, na.rm = T)
#head(tot1)

old <- read.csv("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", skip = 6)
plot(old$Year, old$ATL, type = "l", xlab = "year", ylab = "total catch (pounds)", lwd = 4,
     xlim = c(1980, 2025), ylim = c(7*10^6, 3.3*10^7), bty = "n")
lines(as.numeric(names(tot1)), tot1, lwd = 2, col = 2, lty = 2)
legend("topright", c("previous data", "new data"), col = c(1, 2), lwd = c(4, 2), lty = c(1,
```



9.3 Partition catches by area and sector

Now we will partition out the total catches according to the dolphin MSE areas. The areas are as follows:

- FLK: Florida Keys to Indian River County County (FL)
- NCFL: Brevard (FL) to Southern NC (South of Hatteras)
- NNC: Northern NC (North of Hatteras to NC/VA border)

- VBM: Virginia to Maine

```
# separate out 4 regions --------------------------

d1$region <- "VBM"

# select for FLK
# FL_REG codes: 4: SE FL- Miami-Dade to Indian River County.; 5: NE FL- Brevard to Nassau C
table(d1$FL_REG, d1$NEW_STA, useNA = "always")
```

```
         CT    DE   FLE FLE/GA  FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
  3       0     0     0      0  568     0     0     0     0     0     0     0     0     0
  4       0     0  1022      0    0     0     0     0     0     0     0     0     0     0
  5       0     0   312      0    0     0     0     0     0     0     0     0     0     0
  <NA>   14   114     0   1794    0    54    20   125  1344   108    73    62   644   165

       <NA>
  3       0
  4       0
  5       0
  <NA>    0
```

```
d1$region[which(d1$FL_REG == 3 | d1$FL_REG == 4)] <- "FLK"

# select for NCFL
d1$region[which(d1$FL_REG == 5)] <- "NCFL"
lis <- c("SC", "GA", "FLE/GA")           # NCFL states
d1$region[which(d1$NEW_STA %in% lis)] <- "NCFL"
# NC_REG codes: N: North of Cape Hatteras; S: South of Cape Hatteras
table(d1$NC_REG, d1$NEW_STA, useNA = "always")
```

```
         CT    DE   FLE FLE/GA  FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
         14   114  1334   1794  568    54    20   125   586   108    73    62   644   165
  N       0     0     0      0    0     0     0     0   420     0     0     0     0     0
  S       0     0     0      0    0     0     0     0   338     0     0     0     0     0
  <NA>    0     0     0      0    0     0     0     0     0     0     0     0     0     0
```

```
      <NA>
         0
  N      0
  S      0
  <NA>   0
```

```r
d1$region[which(d1$NC == "S")] <- "NCFL"

# select for NNC
d1$region[which(d1$NC == "N")] <- "NNC"

table(d1$FL_REG, d1$NEW_STA, d1$region)
```

```
, ,  = FLK


     CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  3   0    0    0      0  568    0    0    0    0    0    0    0    0    0
  4   0    0 1022      0    0    0    0    0    0    0    0    0    0    0
  5   0    0    0      0    0    0    0    0    0    0    0    0    0    0

, ,  = NCFL


     CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  3   0    0    0      0    0    0    0    0    0    0    0    0    0    0
  4   0    0    0      0    0    0    0    0    0    0    0    0    0    0
  5   0    0  312      0    0    0    0    0    0    0    0    0    0    0

, ,  = NNC


     CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  3   0    0    0      0    0    0    0    0    0    0    0    0    0    0
  4   0    0    0      0    0    0    0    0    0    0    0    0    0    0
  5   0    0    0      0    0    0    0    0    0    0    0    0    0    0

, ,  = VBM
```

```
     CT    DE   FLE FLE/GA   FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
  3   0     0     0      0     0     0     0     0     0     0     0     0     0     0
  4   0     0     0      0     0     0     0     0     0     0     0     0     0     0
  5   0     0     0      0     0     0     0     0     0     0     0     0     0     0
```

```r
table(d1$NC_REG, d1$NEW_STA, d1$region)  # check classification
```

```
, ,  = FLK


     CT    DE   FLE FLE/GA   FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
      0     0  1022      0   568     0     0     0     0     0     0     0     0     0
  N   0     0     0      0     0     0     0     0     0     0     0     0     0     0
  S   0     0     0      0     0     0     0     0     0     0     0     0     0     0

, ,  = NCFL


     CT    DE   FLE FLE/GA   FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
      0     0   312   1794     0    54     0     0     0     0     0     0   644     0
  N   0     0     0      0     0     0     0     0     0     0     0     0     0     0
  S   0     0     0      0     0     0     0     0   338     0     0     0     0     0

, ,  = NNC


     CT    DE   FLE FLE/GA   FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
      0     0     0      0     0     0     0     0     0     0     0     0     0     0
  N   0     0     0      0     0     0     0     0   420     0     0     0     0     0
  S   0     0     0      0     0     0     0     0     0     0     0     0     0     0

, ,  = VBM


     CT    DE   FLE FLE/GA   FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
     14   114     0      0     0     0    20   125   586   108    73    62     0   165
  N   0     0     0      0     0     0     0     0     0     0     0     0     0     0
  S   0     0     0      0     0     0     0     0     0     0     0     0     0     0
```

Now we will separate out the catches by recreational sector and plot the totals by region and sector.

```r
# separate out by for hire vs. private rec --------------------------
d1$sector <- "ForHire"

d1$sector[which(d1$NEW_MODEN == "Shore")] <- "Rec"
d1$sector[which(d1$NEW_MODEN == "Priv")]  <- "Rec"

table(d1$sector, d1$NEW_MODEN, useNA = "always")    # check classification
```

```
          Cbt Cbt/Hbt  Hbt Priv Shore <NA>
  ForHire 1616     119 2912    0     0    0
  Rec        0       0    0 1750    22    0
  <NA>       0       0    0    0     0    0
```

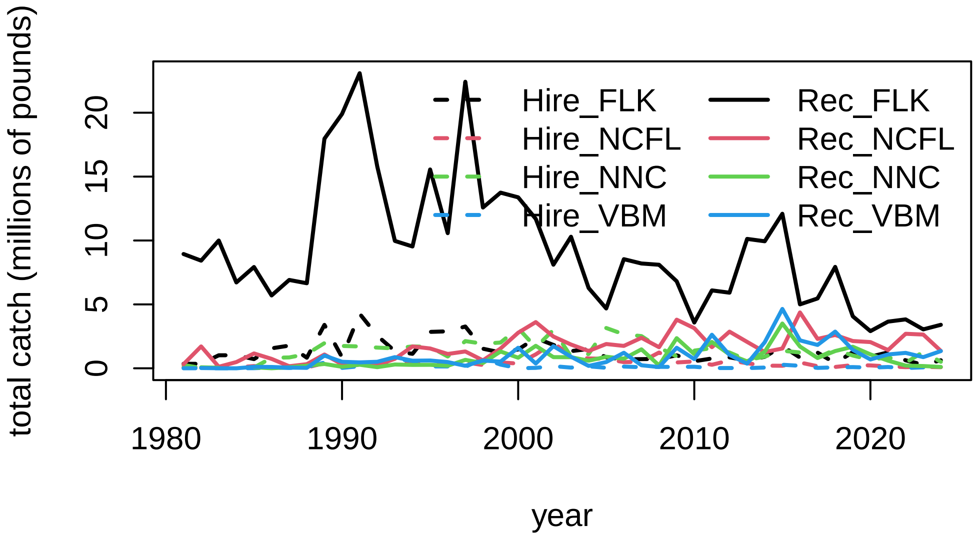
```r
tot <- tapply(d1$LBSEST_SECWWT, list(d1$YEAR, d1$region, d1$sector), sum, na.rm = T)

head(tot)
```

```
, , ForHire

           FLK      NCFL       NNC      VBM
1981  354639.5  74702.50 256755.24  1400.247
1982  347458.2  92271.42  27578.44  2749.224
1983 1020005.8  41784.04  88690.53  5379.177
1984 1021086.8  74658.81        NA  4203.048
1985  717523.0 172989.65  58514.13  8880.857
1986 1562354.0  63555.89 817192.26 61658.193

, , Rec

          FLK      NCFL       NNC        VBM
1981  8942641  322505.7        NA         NA
1982  8422673 1711834.7 78367.161  18727.290
1983 10000662  119370.4 44226.032   5456.237
1984  6717044  486079.7        NA      0.000
1985  7925301 1162708.4 77477.096 112944.274
1986  5699311  736872.0  2549.586 103945.134
```

```r
fin <- data.frame(cbind(tot[, , 1], tot[, , 2]))
names(fin) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM",
                "Rec_FLK", "Rec_NCFL", "Rec_NNC", "Rec_VBM")
head(fin)
```

```
      Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM  Rec_FLK  Rec_NCFL   Rec_NNC
1981  354639.5  74702.50 256755.24  1400.247  8942641  322505.7        NA
1982  347458.2  92271.42  27578.44  2749.224  8422673 1711834.7 78367.161
1983 1020005.8  41784.04  88690.53  5379.177 10000662  119370.4 44226.032
1984 1021086.8  74658.81        NA  4203.048  6717044  486079.7        NA
1985  717523.0 172989.65  58514.13  8880.857  7925301 1162708.4 77477.096
1986 1562354.0  63555.89 817192.26 61658.193  5699311  736872.0  2549.586
        Rec_VBM
1981         NA
1982  18727.290
1983   5456.237
1984      0.000
1985 112944.274
1986 103945.134
```

```r
matplot(rownames(fin), fin/10^6, lty = c(rep(2, 4), rep(1, 4)),
        col = rep(1:4, 2), type = "l", lwd = 2,
        xlab = "year", ylab = "total catch (millions of pounds)")
legend("topright", colnames(fin), lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), lwd = 2
```




```r
# check that totals match
cor(rowSums(fin, na.rm = T), tot1)
```

```
[1] 1
```

```r
mean(rowSums(fin, na.rm = T) - tot1)    # very tiny differences
```

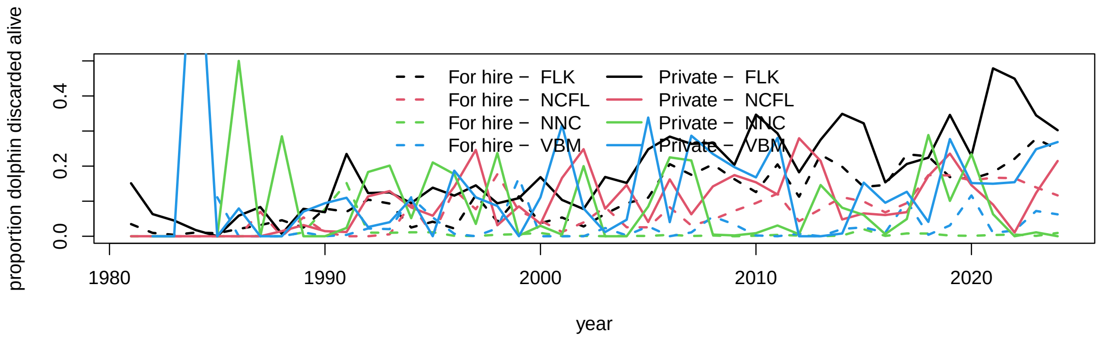
```
[1] 4.233284e-11
```

Here are the catches (in pounds) for years 1981 to 2024.

9.4 Looking at extent of discarding

Discard estimates (known as B2 estimates) are provided in numbers by the MRIP survey but are not available in the Southeast Region Headboat Survey. We can look at the overall numbers of dolphinfish caught versus reported released from the MRIP survey to get an understanding of the extent of discarding.

```
d1$perdis <- d1$B2 / (d1$AB1 + d1$B2)  # percent discarded is B2 (fish released alive) / to

# calculate mean discard rate by year, region, sector ----------------
dis <- tapply(d1$perdis, list(d1$YEAR, d1$region, d1$sector), mean, na.rm = T)
disr <- tapply(d1$perdis, list(d1$region, d1$sector), mean, na.rm = T)

matplot(rownames(dis), cbind(dis[, , 1], dis[, , 2]), lty = c(rep(2, 4), rep(1, 4)), col =
        type = "l", lwd = 2, xlab = "year", ylab = "proportion dolphin discarded alive", yl
legend("top", c(paste("For hire - ", colnames(dis)), paste("Private - ", colnames(dis))),
       lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), lwd = 2, ncol = 2, bty = "n")
```




```
round(disr * 100, 2)  # discarding rates by region and sector
```

```
     ForHire   Rec
FLK    10.79 18.76
NCFL    9.44 11.23
NNC     0.94  9.62
VBM     2.94 12.69
```

According to the MRIP survey and the reported numbers, discarding generally increases over time, excluding some highly anomalous estimates from the early years which are driven by low sample sizes. Discarding is generally higher in the private sector compared to the for-hire sector, and is highest in the Florida Keys region.

9.5 Split the catches by season

The dolphin MSE operates on a seasonal time step and thus the catches and discards must by parsed apart by individual season and year. This is not straightforward because MRIP estimates are reported in two-month waves (Jan - Feb, Mar - Apr, May - Jun, etc.), and the seasons are 3-month periods (Dec - Feb, Mar - May, Jun - Aug, Sep - Nov). For the two waves that have to be split into seasons (e.g., December is lumped with January and February; June is lumped with July and August), we split the wave 50/50%, i.e, we make the assumption that half of the catch in that wave occurs in one season and the other half occurs in the next season.

```
# define seasons ---------------------------
table(d1$WAVE, useNA = "always")
```

```

   0    1    2    3    4    5    6 <NA> 
   5  407  718 1230 1439 1095  581  944 
```

```
d1$YRWAVE <- paste0(d1$YEAR, "_", d1$WAVE)

totw <- tapply(d1$LBSEST_SECWWT, list(d1$YRWAVE, d1$region, d1$sector), sum, na.rm = T)

f <- data.frame(cbind(totw[, , 1], totw[, , 2]))
names(f) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Rec_
head(f)
```

```
          Hire_FLK Hire_NCFL   Hire_NNC Hire_VBM   Rec_FLK  Rec_NCFL Rec_NNC
1981_2          NA        NA         NA       NA 2793285.5 270968.46      NA
1981_3  300873.764        NA 254049.420       NA 3915698.3        NA      NA
1981_4   50017.460        NA   2705.821       NA  905621.2  51537.26      NA
1981_5    3748.301        NA         NA       NA  479834.0        NA      NA
1981_6          NA        NA         NA       NA  848201.5        NA      NA
1981_NA         NA   74702.5         NA 1400.247        NA        NA      NA
        Rec_VBM
1981_2       NA
1981_3       NA
1981_4       NA
1981_5       NA
1981_6       NA
1981_NA      NA
```

```r
f$YEAR <- as.numeric(substr(rownames(f), 1, 4))
f$WAVE <- as.numeric(substr(rownames(f), 6, 8))
head(f)   # table with catches separated by wave and year
```

```
          Hire_FLK Hire_NCFL   Hire_NNC Hire_VBM   Rec_FLK  Rec_NCFL Rec_NNC
1981_2          NA        NA         NA       NA 2793285.5 270968.46      NA
1981_3  300873.764        NA 254049.420       NA 3915698.3        NA      NA
1981_4   50017.460        NA   2705.821       NA  905621.2  51537.26      NA
1981_5    3748.301        NA         NA       NA  479834.0        NA      NA
1981_6          NA        NA         NA       NA  848201.5        NA      NA
1981_NA         NA   74702.5         NA 1400.247        NA        NA      NA
        Rec_VBM YEAR WAVE
1981_2       NA 1981    2
1981_3       NA 1981    3
1981_4       NA 1981    4
1981_5       NA 1981    5
1981_6       NA 1981    6
1981_NA      NA 1981   NA
```

```r
f[is.na(f)] <- 0

f <- f[which(f$YEAR >= 1985), ]
f <- f[which(f$YEAR <= 2024), ]

table(f$YEAR)
```

```

1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000
   7    7    7    7    7    7    7    7    7    7    7    7    6    6    6    6
2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016
   6    6    6    6    6    6    6    6    6    6    6    6    7    7    7    6
2017 2018 2019 2020 2021 2022 2023 2024
   6    6    6    6    6    6    6    6
```

```r
table(f$WAVE)
```

```

 0  1  2  3  4  5  6
15 40 40 40 40 40 40
```

```
# deal with NAs in many locations ---------------
# calculate percentages of catch by wave, for each region and fleet
avw <- tapply(d1$LBSEST_SECWWT, list(d1$WAVE, d1$region, d1$sector), sum, na.rm = T)
avw <- avw[-1, , ]
avw <- data.frame(cbind(avw[, , 1], avw[, , 2]))
names(avw) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Re
avwp <- apply(avw, 2, function (x) x / sum(x, na.rm = T))
#colSums(avwp)

# go through each column, find NAs and fill in waves by percentages
pre_colsum <- colSums(f, na.rm = T)  # column sums to check later

lis <- which(f$WAVE == 0)  # list of rows with NA wave data

for (i in 1:8) {      # loop through each column
  for(j in lis) {     # loop through list of NA waves
    if (f[j, i] > 0) {  # if there are NA wave catches, split by known proportion from othe
      miscatch <- avwp[, i] * f[j, i]   # missing catch is known proportion * NA wave catch
      f[which(f$YEAR == f$YEAR[j] & f$WAVE != 0), i] <- f[which(f$YEAR == f$YEAR[j] & f$WAV
      f[j, i] <- 0   # set to zero after catch is attributed to waves
    }
  }
}
f[lis, 1:8] # should be all zeros now
```

```
        Hire_FLK Hire_NCFL Hire_NNC Hire_VBM Rec_FLK Rec_NCFL Rec_NNC Rec_VBM
1985_NA        0         0        0        0       0        0       0       0
1986_NA        0         0        0        0       0        0       0       0
1987_NA        0         0        0        0       0        0       0       0
1988_NA        0         0        0        0       0        0       0       0
1989_NA        0         0        0        0       0        0       0       0
1990_NA        0         0        0        0       0        0       0       0
1991_NA        0         0        0        0       0        0       0       0
1992_NA        0         0        0        0       0        0       0       0
1993_NA        0         0        0        0       0        0       0       0
1994_NA        0         0        0        0       0        0       0       0
1995_NA        0         0        0        0       0        0       0       0
1996_NA        0         0        0        0       0        0       0       0
2013_0         0         0        0        0       0        0       0       0
2014_0         0         0        0        0       0        0       0       0
```

```
2015_0          0         0         0         0        0         0        0        0
```

```
pos_colsum <- colSums(f, na.rm = T)
pre_colsum[1:8] == pos_colsum[1:8]  # check that catch was parsed correctly - column sums s
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM 
     TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE 
```

```
f <- f[which(f$WAVE != 0), ]  # remove NA waves which now contain no catch

# split waves 3 and 6 50/50% to parse into seasons ------------------
lis <- which(f$WAVE == 3 | f$WAVE == 6)  # list of 3 and 6 waves for all years

for (i in lis) {          # go through list of waves
  temp <- f[i, ] / 2      # split catch in half
  temp[9] <- f$YEAR[i]    # reformat year and wave data columns
  temp[10] <- f$WAVE[i]
  temp <- rbind(temp, temp)   # create two half-waves and relabel with new wave reference n
  temp[1, 10] <- temp[1, 10] - 0.5
  temp[2, 10] <- temp[2, 10] + 0.5
  f <- rbind(f, temp)         # concatenate with main data table
  }

f <- f[-lis, ]  # remove waves that have been split in half

pos_colsum <- colSums(f, na.rm = T)
pre_colsum[1:8] == pos_colsum[1:8]  # check that catch was parsed correctly - column sums s
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM 
     TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE 
```

```
f <- f[order(f$YEAR, f$WAVE), ]  # reorder table in chronological order

f$YEAR[which(f$WAVE == 6.5)] <- f$YEAR[which(f$WAVE == 6.5)] + 1  # add year to December wa
f$WAVE[which(f$WAVE == 6.5)] <- 0.5  # change December wave to small number so it gets summ

table(f$WAVE)
```

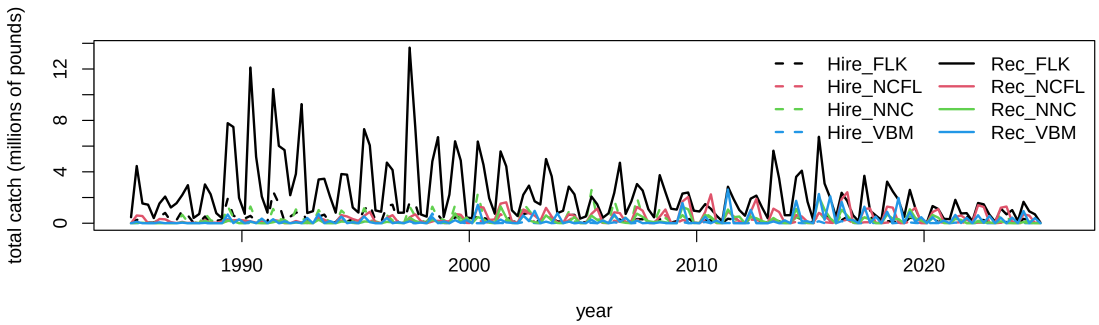
```
0.5   1   2 2.5 3.5   4   5 5.5
 40  40  40  40  40  40  40  40
```

```
f$seas <- cut(f$WAVE, breaks = c(0, 1.5, 3, 4.5, 6))
#table(f$seas)
f$seas1 <- NA
f$seas1[which(f$seas == "(0,1.5]")] <- "DJF"  # Dec Jan Feb is 0.5 and 1
f$seas1[which(f$seas == "(1.5,3]")] <- "MAM"  # Mar Apr May is 2 and 2.5
f$seas1[which(f$seas == "(3,4.5]")] <- "JJA"  # Jun Jul Aug is 3.5 and 4
f$seas1[which(f$seas == "(4.5,6]")] <- "SON"  # Sep Oct Nov is 5 and 5.5

temp <- tapply(f[, 1], list(f$seas1, f$YEAR), sum, na.rm = T)
temp1 <- matrix(temp)
for (i in 2:8) {
  temp2 <- matrix(tapply(f[, i], list(f$seas1, f$YEAR), sum, na.rm = T))
  temp1 <- cbind(temp1, temp2)
  }

findat <- data.frame(sort(rep(as.numeric(colnames(temp)), 4)),
                     rep(rownames(temp), ncol(temp)), temp1)
names(findat) <- c("year", "season", names(f)[1:8])

colSums(f[, 1:8], na.rm = T) == colSums(findat[, 3:10], na.rm = T)  # check that total catc
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM
     TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE
```

We plot the results by region and sector, over the time series of year and season. We can clearly see the seasonality in the catch by region.

```
findat$yrseas <- findat$year + as.numeric(as.factor(findat$season))/4-0.125
matplot(findat$yrseas, findat[, 3:10]/10^6, lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2
        xlab = "year", ylab = "total catch (millions of pounds)")
legend("topright", colnames(findat[, 3:10]), lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4,
```




```
# check that totals match
cbind(colSums(findat[, 3:10], na.rm = T), colSums(fin[-c(1:5), ], na.rm = T))
```

```
                [,1]      [,2]
Hire_FLK    55604999  54887476
Hire_NCFL   16909916  16736926
Hire_NNC    57944903  57886388
Hire_VBM     5836953   5828072
Rec_FLK    371462933 363537632
Rec_NCFL    70803866  69641157
Rec_NNC     32556596  32479119
Rec_VBM     40081529  39968585
```

```
# the values should be close but slightly >1 because the last data set includes December 19
colSums(findat[, 3:10], na.rm = T) / colSums(fin[-c(1:5), ], na.rm = T)
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM
 1.013073  1.010336  1.001011  1.001524  1.021800  1.016696  1.002385  1.002826
```

Here we will format the data for input to the operating model.

```
tab3 <- findat

flt <- c(rep("Hire", nrow(tab3)*4), rep("Rec", nrow(tab3)*4))

yrmon <- as.numeric(tab3$yrseas)
yrs <- floor(yrmon)
```

```r
qrt <- (yrmon - floor(yrmon)) * 4 + 0.5
area <- c(rep("FLK", nrow(tab3)), rep("NCFL", nrow(tab3)),
          rep("NNC", nrow(tab3)), rep("VBM", nrow(tab3)))
area <- rep(area, 2)

findat1 <- data.frame(yrs, qrt, flt, area, matrix(as.matrix(tab3[,3:10])))
names(findat1) <- c("Year", "Quarter", "Fleet", "Area", "Catch_lbs")
head(findat1)
```

```
  Year Quarter Fleet Area Catch_lbs
1 1985       1  Hire  FLK   4462.09
2 1985       2  Hire  FLK 253487.38
3 1985       3  Hire  FLK 380811.37
4 1985       4  Hire  FLK  41469.99
5 1986       1  Hire  FLK  40545.41
6 1986       2  Hire  FLK 501184.61
```

```r
#plot(findat1$Catch_lbs, type = "l")
#plot(findat1$Catch_lbs ~ factor(findat1$Area))
#plot(findat1$Catch_lbs ~ factor(findat1$Quarter))
#plot(findat1$Catch_lbs ~ factor(findat1$Year))

findat1 <- findat1[which(findat1$Year <= 2022), ]
findat1 <- findat1[which(findat1$Year >= 1986), ]
#apply(findat1, 2, table)

findat <- findat[which(findat$year >=1986 & findat$year <= 2022), ]
#check that the numbers are exactly the same
tapply(findat1$Catch_lbs, list(findat1$Area, findat1$Fleet), sum, na.rm = T)
```

```
         Hire       Rec
FLK  54012811 357091651
NCFL 16532764  65643193
NNC  56097190  32197622
VBM   5736592  37749487
```

```r
colSums(findat[3:10], na.rm = T)
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM
 54012811  16532764  56097190   5736592 357091651  65643193  32197622  37749487
```

```
# output final numbers in correct format
write.csv(findat1, file = "data/FINAL_files/rec_catch_TomFormat.csv", row.names = FALSE)
```

10 NOAA Fisheries commercial data

We do not display the commercial data analysis here because it is based on logbook reporting and trip ticket databases, which both contain confidential data. The code for processing the commercial data can be seen in the file "commercial.R" and the steps are outlined below.

10.1 Analysis of logbook data

We first use the pelagic longline logbook data to understand how the fleet operates in the high seas on a seasonal basis. These numbers are used later to estimate the seasonality of the international catches, because those are only reported on an annual basis but quarterly values are needed.

The steps are as follows:

1. Read in the pelagic longline data which contains set level entries including location of fishing, target species, gear used, other details of the fishing trip and number of dolphin caught, discarded alive, discarded dead, and the total pounds of dolphin kept.
2. Format the longitude and latitude of the sets into decimal degrees, and find the points that fall within each of the seven polygons representing the areas of the operating model. Check that subsetting was done correctly and add and area identifier.
3. Look at the characteristics of the dolphin catch and calculate the discard rates by trip. Dead discarding rate is found to be 1.4% on average and total discard rate is 2.9% on average.
4. Standardize date formatting and add identifier for season (quarter). Specify a new year variable to allow December to be grouped with January and February.
5. Summarize the total dolphin catch (in pounds) by year, quarter, and area.

6. Convert the total pounds by year-quarter and area to proportions such that each year-quarter combination has four seasonal proportions summing to one. Save the matrix of proportions for later analysis.

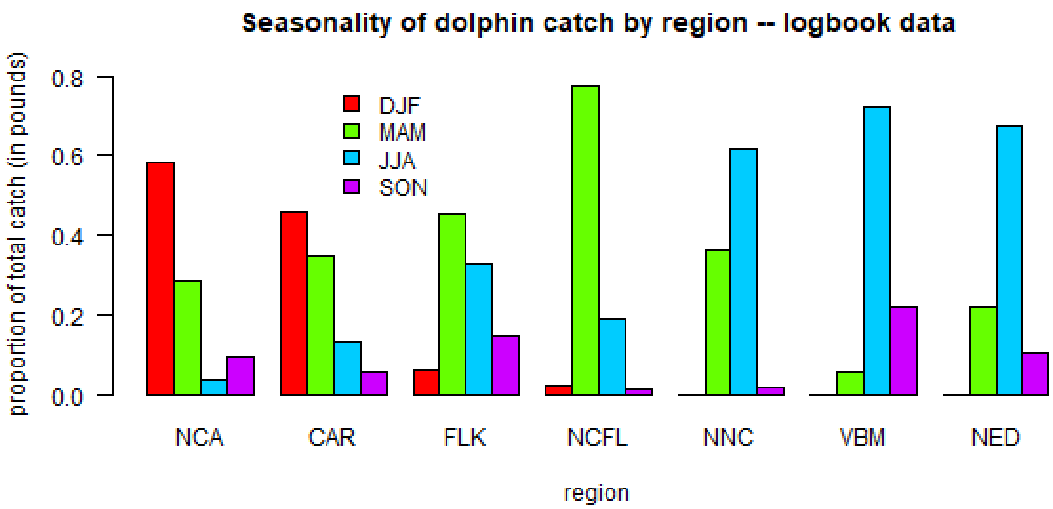


Figure 10.1: Seasonality of the pelagic longline fleet according to the logbook data. Figure shows the proportion of dolphin catch occurring in each season, for each region of the operating model.

10.2 Analysis of trip ticket data

Not all of the commercial dolphin catch activity is reported in the logbooks because dolphin are targeted via a variety of permits and gear types, not all of which have the same reporting requirements. Therefore, in order to parse the commercial catches by year, area and quarter, we use the commercial trip ticket data compiled by NOAA Fisheries. These are the same data used for quota monitoring and thus are the best representation of the total catch.

The steps are as follows:

1. Read in the trip ticket data file, which contains commercial dolphinfish landings for the Atlantic coast from 1986-2023, including all landings by year even if the

gear type or area is unknown. The dataset specifies the year and month of landing, an area of fishing, gear, and the state where landed. Pounds are in units of whole weight.

2. Read in operating model regions and classify the list of logbook areas into those seven regions. Assign each reported area to an operating model region accordingly.
3. Standardize date formatting and add identifier for season (quarter). Specify a new year variable to allow December to be grouped with January and February.
4. Summarize the landings by year-quarter, gear (PLL gear vs. other gears) and region.
5. For most combinations of year-quarter, region and gear, there are substantial landings for which the area of fishing is unknown and therefore the region cannot be defined. We assume that those landings follow the same distribution with regard to region as the other landings for that year-quarter and gear combination. For each year-quarter/gear combination, we calculate the proportion of catch that occurs in each of the seven regions. The total landings of unknown regions are multiplied by those proportions and then added to the landings from known regions. In cases where none of the areas are known for that year-quarter/gear combination, proportions from the quarter immediately previous to that quarter are used. Sum the landings by gear type to get the total landings for each year-quarter and region.
6. Reformat the data according to the specifications needed for input to the operating model (columns specifying the year, quarter, fleet, area and total catch).
7. Sum landings by year from the original data file and the final outputs to ensure that data were processed correctly and no catch is missing.

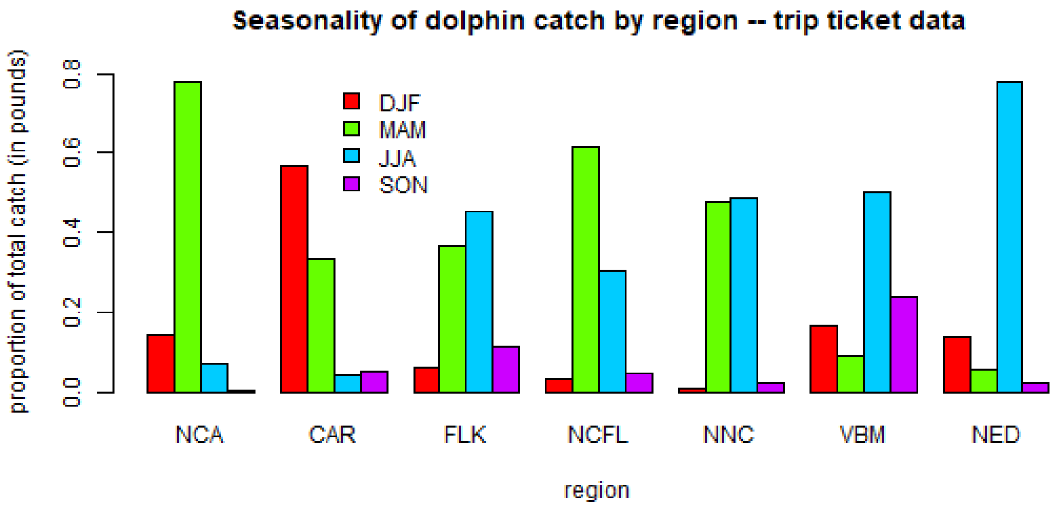


Figure 10.2: Seasonality of the commercial dolphin catch according to the trip ticket data. Figure shows the proportion of dolphin catch occurring in each season, for each region of the operating model.

11 Analysis of international catches in the high seas

Here we will explore commercial and recreational catch data as estimated by the Sea Around Us Project. This data set includes reported data for fisheries worldwide as well as reconstructed estimates for unreported catches. For each country, project scientists assemble available data on catch and use additional sociological and fishing data plus expert information and opinion, to produce their best estimates of total catch from all fishing sectors and for all species globally.

11.1 Data access and upload

The high seas data are accessed in the Basic Search High seas section of the site. The individual regions can be clicked on (e.g., Atlantic, Western Central) and then the data are downloaded using the Download Data button.

The current version of data in use is version 50.1, accessed June 2024 (the data files can be found on github in the SEFSC-dolphin-analyses/data /SAU/ folder). This was still the most recent version of data available as of September 2025.

11.2 Read in the high seas catches

First we read in the high seas catches for the Western Central Atlantic and Northwest Atlantic zones. These are equivalent to FAO areas 21 and 31.

```
# clear workspace
rm(list = ls())

if(!require("pals")) install.packages("pals")
library(pals)
```

```
# find data files
hslis <- dir("data/SAU/")[grep("HighSeas", dir("data/SAU/"))]

# identify areas 21 and 31
hslis[1]; hslis[3]
```

```
[1] "SAU HighSeas 21 v50-1.csv"
```

```
[1] "SAU HighSeas 31 v50-1.csv"
```

```
#download data
h1 <- read.csv(paste0("data/SAU/", hslis[1]))
h3 <- read.csv(paste0("data/SAU/", hslis[3]))

# confirm correct areas
#unique(h1$area_name)
#unique(h3$area_name)

hs <- rbind(h1, h3)
```

We extract the data for dolphinfish and look at the characteristics of the catches. Most of the catches are in the Western Central Atlantic, and the primary fishing entities are Venezuela and Saint Lucia. The fishing sector is all industrial, and the vast majority of the catch is reported landings, with only a tiny fraction of discards or unreported landings listed in the database. The gear type is primarily long line.

```
# check species name and filter out dolphinfish
unique(hs$common_name)[grep("dolphin", unique(hs$common_name))]
```

```
[1] "Common dolphinfish"
```

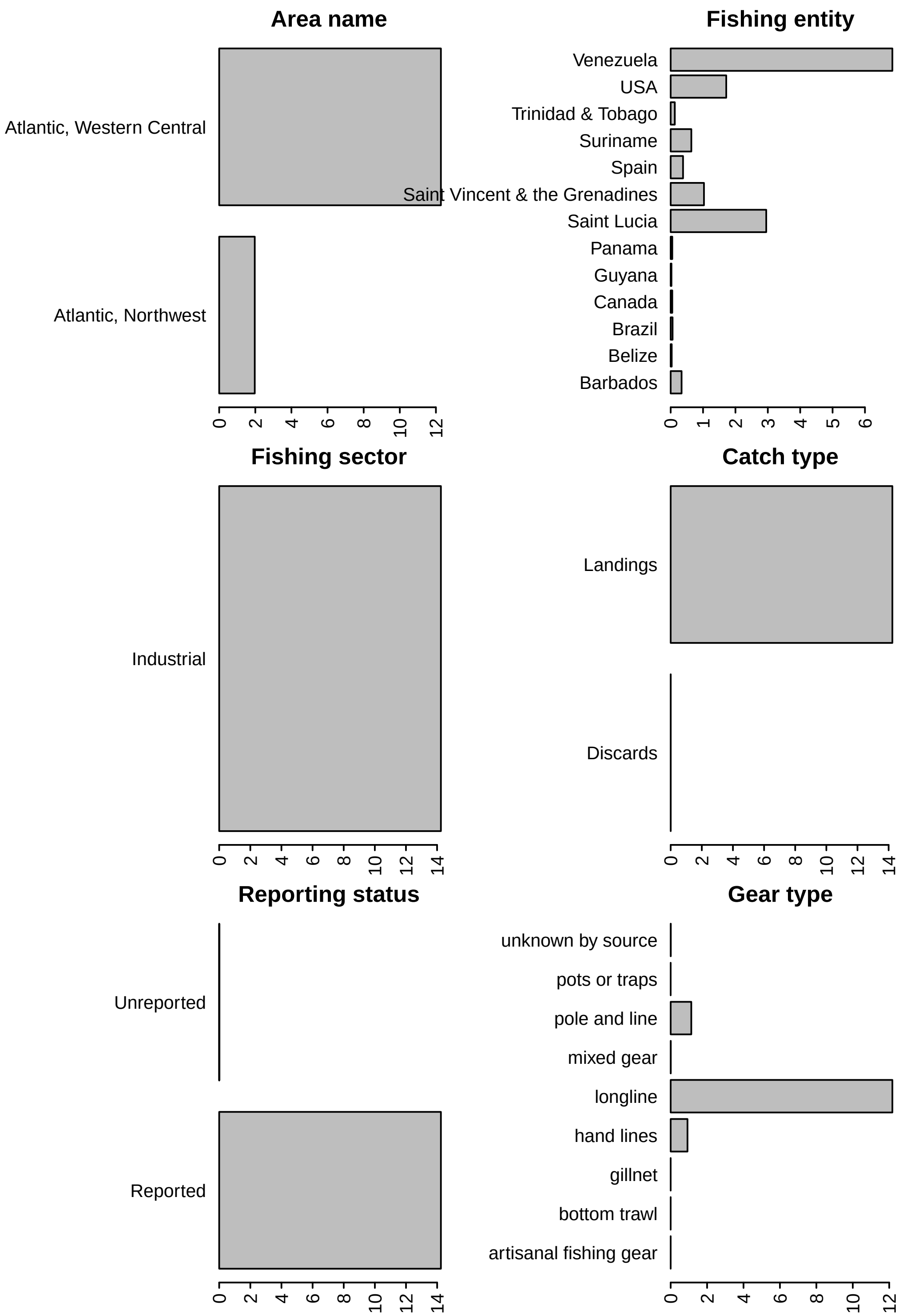
```
hs <- hs[which(hs$common_name == "Common dolphinfish"), ]

# look at the categories within each column of data
apply(hs[1:13], 2, table)
```

```
$area_name

      Atlantic, Northwest Atlantic, Western Central
                      113                       160

$area_type

high_seas
      273

$year

1985 1986 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012
   1    1    1    1    1    1    1    6    4    1    5   10    5    8   10   12
2013 2014 2015 2016 2017 2018 2019
  24   22   30   33   33   29   34

$scientific_name

Coryphaena hippurus
                273

$common_name

Common dolphinfish
               273

$functional_group

Large pelagics (>=90 cm)
                     273

$commercial_group

Perch-likes
        273

$fishing_entity

                      Barbados                          Belize
                             6                               3
```

```
                  Brazil                          Canada
                      19                              18
                  Guyana                          Panama
                       1                               1
             Saint Lucia Saint Vincent & the Grenadines
                       5                              28
                   Spain                        Suriname
                      80                               1
       Trinidad & Tobago                             USA
                      14                              74
               Venezuela
                      23

$fishing_sector

Industrial
       273

$catch_type

Discards Landings
       5      268

$reporting_status

  Reported Unreported
       253         20

$gear_type

artisanal fishing gear           bottom trawl                gillnet
                     7                      9                     11
            hand lines               longline             mixed gear
                    17                    187                      1
         pole and line          pots or traps      unknown by source
                    19                      9                     13

$end_use_type

                        Direct human consumption    Fishmeal and fish oil
                       5                     194                       37
```

```
Other
   37
```

```
# convert from metric tonnes to pounds
hs$pounds <- hs$tonnes * 1000 * 2.20462

# produce barplot of key data fields
par(mar = c(3, 20, 3, 1), mfrow = c(3, 2), mex = 0.5)
barplot(tapply(hs$pounds/10^6, hs$area_name, sum, na.rm = T), las = 2, xlab = "millions of
#barplot(tapply(hs$pounds/10^6, hs$area_type, sum, na.rm = T), las = 2, ylab = "millions of
#barplot(tapply(hs$pounds/10^6, hs$year, sum, na.rm = T), las = 2, ylab = "millions of poun
barplot(tapply(hs$pounds/10^6, hs$fishing_entity, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(hs$pounds/10^6, hs$fishing_sector, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(hs$pounds/10^6, hs$catch_type, sum, na.rm = T), las = 2, xlab = "millions of
barplot(tapply(hs$pounds/10^6, hs$reporting_status, sum, na.rm = T), las = 2, xlab = "milli
barplot(tapply(hs$pounds/10^6, hs$gear_type, sum, na.rm = T), las = 2, xlab = "millions of
```


Area name
Atlantic, Western Central
Atlantic, Northwest
0 2 4 6 8 10 12
Fishing entity
Venezuela
USA
Trinidad & Tobago
Suriname
Spain
Saint Vincent & the Grenadines
Saint Lucia
Panama
Guyana
Canada
Brazil
Belize
Barbados
0 1 2 3 4 5 6
Fishing sector
Industrial
0 2 4 6 8 10 12 14
Catch type
Landings
Discards
0 2 4 6 8 10 12 14
Reporting status
Unreported
Reported
0 2 4 6 8 10 12 14
Gear type
unknown by source
pots or traps
pole and line
mixed gear
longline
hand lines
gillnet
bottom trawl
artisanal fishing gear
0 2 4 6 8 10 12

```
round(tapply(hs$pounds, hs$catch_type, sum, na.rm = T) / sum(hs$pounds) * 100, 2)   # << 1%
```

```
Discards Landings
    0.03    99.97
```

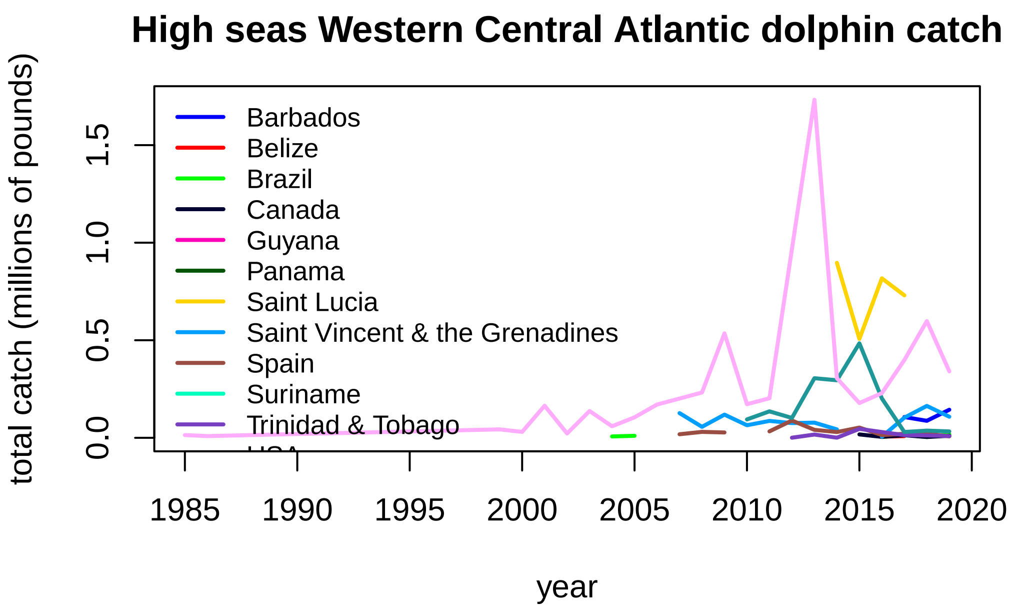
```
round(tapply(hs$pounds, hs$reporting_status, sum, na.rm = T) / sum(hs$pounds) * 100, 2)  #
```

```
  Reported Unreported
     99.97       0.03
```

11.3 Summarize the high seas catches

We plot the annual catches by fishing entity to look at trends over time.

```
hs$fishing_entity <- as.factor(hs$fishing_entity)
hs$fishing_entity <- droplevels(hs$fishing_entity)

par(mar = c(5, 5, 3, 1), mfrow = c(1, 1))
tab <- tapply(hs$pounds, list(hs$year, hs$fishing_entity), sum, na.rm = T)
matplot(as.numeric(rownames(tab)), tab/10^6, type = "l", col = glasbey(ncol(tab)), lty = 1,
        main = "High seas Western Central Atlantic dolphin catch")
legend("topleft", colnames(tab), col = glasbey(ncol(tab)), lty = 1, lwd = 2, cex = 0.8, bty
```



Finally we output the subset of high seas dolphin catch for the two areas. Because we have already accounted for the USA catches in our national reporting and are aiming to only summarize international catches here, we filter out the USA catches from the Sea Around Us database.

```
hs <- hs[which(hs$fishing_entity != "USA"), ]

hs$year <- factor(hs$year, levels = 1950:2019)
tab <- tapply(hs$pounds, list(hs$year, hs$area_name), sum, na.rm = T)
tab[is.na(tab)] <- 0
tail(tab, 40)
```

```
     Atlantic, Northwest Atlantic, Western Central
1980               0.000                     0.000
1981               0.000                     0.000
1982               0.000                     0.000
1983               0.000                     0.000
1984               0.000                     0.000
1985               0.000                 14037.129
1986               0.000                  8748.399
1987               0.000                     0.000
```

```
1988             0.000           0.000
1989             0.000           0.000
1990             0.000           0.000
1991             0.000           0.000
1992             0.000           0.000
1993             0.000           0.000
1994             0.000           0.000
1995             0.000           0.000
1996             0.000           0.000
1997             0.000           0.000
1998             0.000           0.000
1999             0.000       43168.296
2000             0.000       30284.627
2001             0.000      164272.223
2002             0.000       22147.477
2003             0.000      137103.181
2004             0.000       66058.258
2005             0.000      115171.863
2006             0.000      170522.101
2007         14687.527      330838.620
2008         28552.636      340863.719
2009         18719.121      662720.661
2010          4510.085      232093.837
2011         21589.424      300498.224
2012         72398.812     1059210.392
2013         34741.604     2471791.396
2014         27922.953     1247462.983
2015         90313.638      710059.125
2016         15319.433     1094893.891
2017         34806.933     1374259.739
2018         29328.358      885814.754
2019         21870.621      632747.157
```

```
# write summary to merge with EEZ data
write.csv(tab, file = "data/SAU/high_seas_by_year.csv")
```

11.4 Distribute the high seas catches across quarters of the year

The international catches are only reported as annual catches and do not have any month or season associated with them. However, the MSE operating model requires seasonal catches. The majority of the international catches are caught by longline, so we will assume that the fleet distribution of the U.S. pelagic longline is representative of the seasonality of the international high seas fleets. Now we parse the annual catches according to the seasonality of the U.S. fleet.

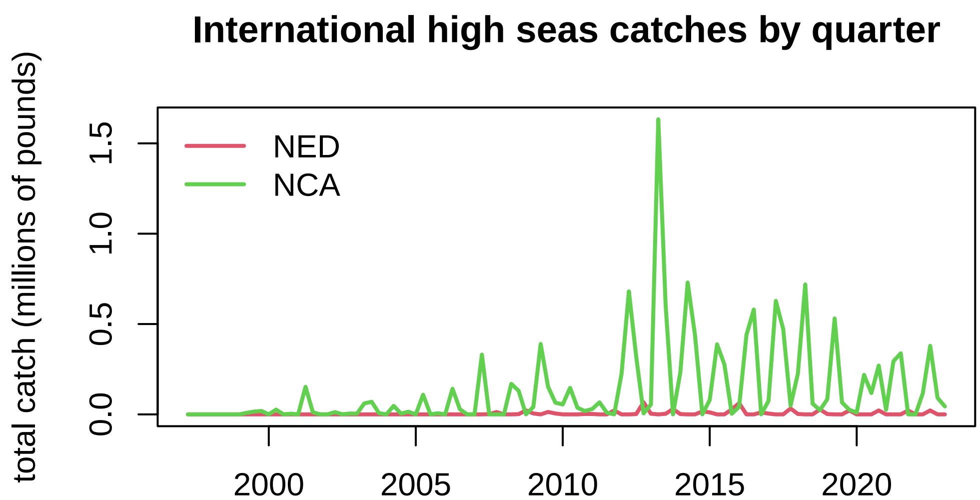
At the time of analysis, the SAU database only contained catches to 2019, so we interpolated the 2019 values to cover years 2020 - 2022.

```
# read in the percentage of the catch by area and year-quarter combinations from the U.S. P
load("data/outputs/per_PLLcatch_by_area_yearquarter.RData")

tab <- tab[which(rownames(tab) >= 1997), ]  # PLL data are only available from 1997 on
#tab[nrow(tab), ]
tab <- rbind(tab, tab[nrow(tab), ], tab[nrow(tab), ], tab[nrow(tab), ])  # interpolate 2019
tail(tab)
```

```
     Atlantic, Northwest Atlantic, Western Central
2017            34806.93                 1374259.7
2018            29328.36                  885814.8
2019            21870.62                  632747.2
                21870.62                  632747.2
                21870.62                  632747.2
                21870.62                  632747.2
```

```
NED <- percatch[, , 1]   # create matrices for final catch data
NCA <- percatch[, , 1]

for (i in 1:ncol(percatch))  {
  NED[, i] <- percatch[, i, 7] * tab[i, 1]   # these are the percentages for the NED (Atlan
  NCA[, i] <- percatch[, i, 1] * tab[i, 2]   # these are the percentages for the NCA (Weste
  }

# plot the parsed out data by year-quarter
yrs <- sort(rep(as.numeric(colnames(NCA)), 4))
qrt <- rep(1:4, length(yrs)/4)
```

```r
matplot(yrs + qrt/4, cbind(matrix(NED), matrix(NCA))/10^6, lty = 1, lwd = 2,
        col = 2:3, type = "l", main = "International high seas catches by quarter",
        xlab = "", ylab = "total catch (millions of pounds)")
legend("topleft", c("NED", "NCA"), lwd = 2, col = 2:3, bty = "n")
```




```r
# save NED data in this format because it needs to be combined with Canadian EEZ catches
save(NED, file = "data/outputs/NED_year_quarter.RData")
```

Finally, we format the data in the format for input into the operating model. We only include NCA here, as the NED high seas catches must be combined with international catches in territorial waters of Canada and France in the NED region.

```r
# format for operating model
flt <- rep("Intl", length(yrs))
are <- rep("NCA", length(yrs))
findat <- data.frame(cbind(yrs, qrt, flt, are, matrix(NCA)))
names(findat) <- c("Year", "Quarter", "Fleet", "Area", "Catch_lbs")
findat$Catch_lbs <- as.numeric(findat$Catch_lbs)
head(findat)
```

```
  Year Quarter Fleet Area Catch_lbs
1 1997       1  Intl  NCA         0
2 1997       2  Intl  NCA         0
3 1997       3  Intl  NCA         0
4 1997       4  Intl  NCA         0
5 1998       1  Intl  NCA         0
6 1998       2  Intl  NCA         0
```

```
#plot(findat$Catch_lbs, type = "l")
#plot(findat$Catch_lbs ~ factor(findat$Area))
#plot(findat$Catch_lbs ~ factor(findat$Quarter))
#plot(findat$Catch_lbs ~ factor(findat$Year))
#apply(findat, 2, table)
#barplot(tapply(findat$Catch_lbs, list(findat$Quarter, findat$Area), sum, na.rm = T), besid

write.csv(findat, file = "data/FINAL_files/NCA_Intl_TomFormat.csv", row.names = FALSE)
```

12 Analysis of international catches in territorial waters

12.1 Data access and upload

For catches within each country's exclusive economic zones, data are downloaded from the Sea Around Us site (Pauly, Zeller, and Palomares, 2020.). In the Biodiversity by Taxon Tools and Data, we filter by EEZ for All regions, and then filter by Taxon at the Species level for Common dolphinfish (Coryphaena hippurus). After clicking "View graphs" one can access the full data set using the "Download Data" button.

12.2 Read in the catches within excluzive economic zones (EEZs)

Now we read in the catches reported by each country or jurisdiction's respective EEZ. Because we have extracted dolphin catches for EEZs worldwide, we need to separate out the catches for the Atlantic Ocean basin.

```
# load libraries
library(pals)

rm(list = ls())
d <- read.csv("data/SAU/SAU Taxa 600006 v50-1.csv", header = T, sep = ",")  # v. 50.1 - Jun
head(d)
```

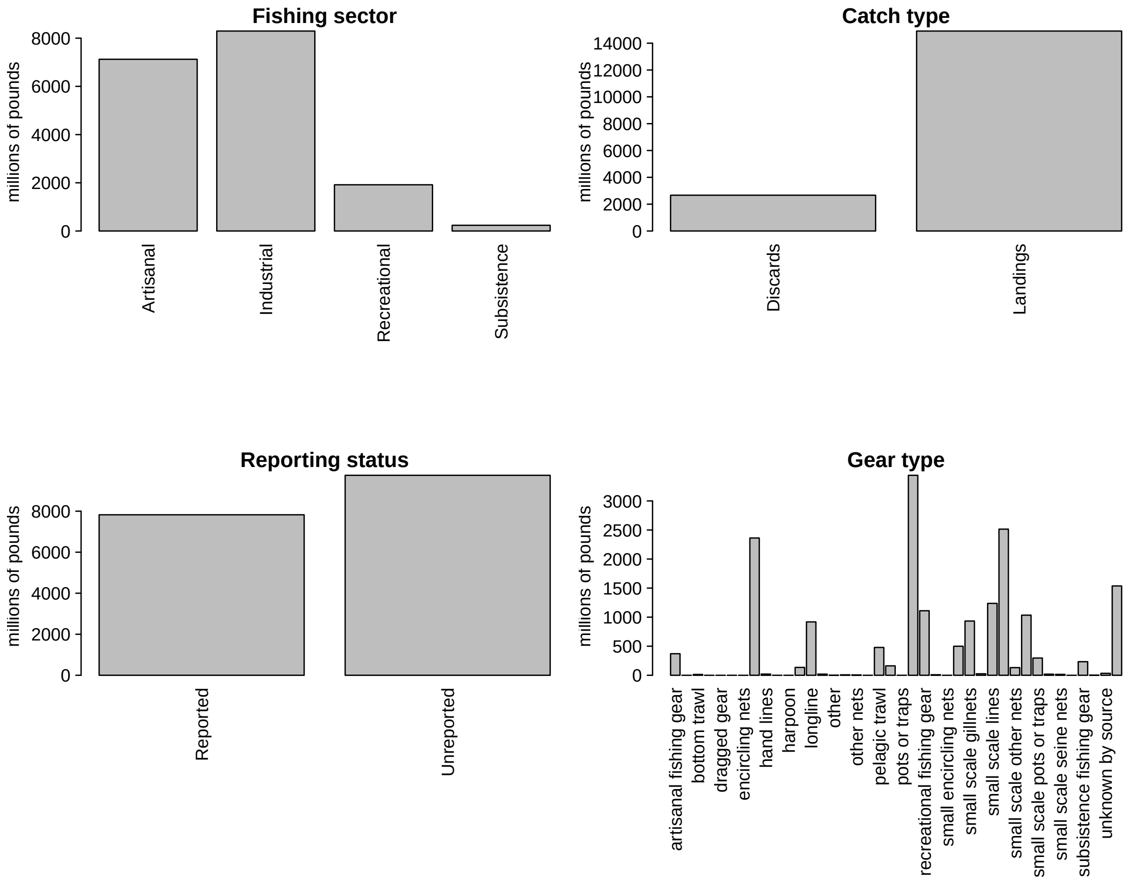
```
    area_name area_type year     scientific_name        common_name
1   Australia       eez 1950 Coryphaena hippurus Common dolphinfish
2   Australia       eez 1950 Coryphaena hippurus Common dolphinfish
3   Australia       eez 1950 Coryphaena hippurus Common dolphinfish
4     Bahamas       eez 1950 Coryphaena hippurus Common dolphinfish
5    Barbados       eez 1950 Coryphaena hippurus Common dolphinfish
```

```
6 Bermuda (UK)       eez 1950 Coryphaena hippurus Common dolphinfish
          functional_group commercial_group fishing_entity fishing_sector
1 Large pelagics (>=90 cm)      Perch-likes      Australia     Industrial
2 Large pelagics (>=90 cm)      Perch-likes      Australia     Industrial
3 Large pelagics (>=90 cm)      Perch-likes      Australia     Industrial
4 Large pelagics (>=90 cm)      Perch-likes        Bahamas   Recreational
5 Large pelagics (>=90 cm)      Perch-likes       Barbados   Recreational
6 Large pelagics (>=90 cm)      Perch-likes   Bermuda (UK)      Artisanal
  catch_type reporting_status                 gear_type
1   Landings         Reported                  longline
2   Landings         Reported                  longline
3   Landings         Reported                  longline
4   Landings       Unreported recreational fishing gear
5   Landings       Unreported recreational fishing gear
6   Landings         Reported         small scale lines
              end_use_type       tonnes landed_value
1    Fishmeal and fish oil  0.003373202 3.438170e-05
2 Direct human consumption  6.739657104 1.040254e+04
3                    Other  0.003373202 3.438170e-05
4 Direct human consumption 48.665105143 8.262728e+00
5 Direct human consumption  0.009158559 4.846886e-04
6 Direct human consumption  1.511085514 3.199403e+03
```

```r
#dim(d)
#apply(d[1:13], 2, table)

# convert tonnes to pounds
d$lbs <- d$tonnes * 1000 * 2.20462
```

We look at the characteristics of the catches within EEZs. These are largely industrial and artisanal in nature. Landings are the majority of the catch but discards are also present. There are more unreported landings than reported landings and a variety of gear types are used.

```r
# look at characteristics of data
par(mfrow = c(2, 2), mar = c(20, 7, 3, 1), mex = 0.5, mgp = c(5.5, 1, 0))
barplot(tapply(d$lbs, d$fishing_sector, sum, na.rm = T)/10^6, las = 2, ylab = "millions of
barplot(tapply(d$lbs, d$catch_type, sum, na.rm = T)/10^6, las = 2, ylab = "millions of poun
barplot(tapply(d$lbs, d$reporting_status, sum, na.rm = T)/10^6, las = 2, ylab = "millions o
barplot(tapply(d$lbs, d$gear_type, sum, na.rm = T)/10^6, las = 2, ylab = "millions of pound
```




```
round(tapply(d$lbs, d$catch_type, sum, na.rm = T) / sum(d$lbs) * 100, 2)    # 15% are discar
```

```
Discards Landings
   15.18    84.82
```

```
round(tapply(d$lbs, d$reporting_status, sum, na.rm = T) / sum(d$lbs) * 100, 2)   # 55% are u
```

```
  Reported Unreported
     44.52      55.48
```

12.3 Separating the EEZs by area

Now we have to manually assign the different EEZs to respective areas. We separate the geographical areas within the Atlantic Ocean (North and South) and lump all other EEZs into "other oceans." This latter category is largely the Pacific Ocean and to a lesser extent the Indian Ocean. We remove the high seas catch from this database as it is not ocean-specific and we have extracted the ocean-specific high seas catch previously.

```
# remove the high seas catch from this database
d <- d[which(d$area_type != "high_seas"), ]
d$fishing_entity <- as.factor(d$fishing_entity)
d$fishing_entity <- droplevels(d$fishing_entity)
table(d$area_type)  # this should be all "eez"
```

```
  eez
63056
```

```
# Western Central Atlantic EEZs
WCA <- c("Bahamas", "Barbados", "Cayman Isl. (UK)", "Dominica", "Dominican Republic", "Gren
         "Aruba (Netherlands)", "Haiti", "Guadeloupe  (France)", "Jamaica", "Martinique (Fr
         "Montserrat (UK)", "Nicaragua (Caribbean)", "Puerto Rico (USA)", "Saint Kitts & Ne
         "Saint Lucia", "Saint Vincent & the Grenadines", "Trinidad & Tobago", "US Virgin I
         "St Barthelemy (France)", "St Martin (France)", "Curaçao (Netherlands)", "Bonaire 
         "Saba and Sint Eustatius (Netherlands)", "Sint Maarten (Netherlands)", "Honduras (
         "Colombia (Caribbean)", "Mexico (Atlantic)", "Turks & Caicos Isl. (UK)", "Guyana",
         "Antigua & Barbuda", "Belize", "British Virgin Isl. (UK)", "French Guiana", "Cuba"
         "Anguilla (UK)", "Suriname", "Costa Rica (Caribbean)", "Guatemala (Caribbean)",
         "Panama (Caribbean)", "Bermuda (UK)", "CuraÃ§ao (Netherlands)")

# Brazil
Bra <- c("Brazil (mainland)", "Fernando de Noronha (Brazil)",
         "St Paul and St. Peter Archipelago (Brazil)")

# Canada
Can <-  c("Canada (East Coast)", "Saint Pierre & Miquelon (France)")

# United States
USA <-  c("USA (East Coast)", "USA (Gulf of Mexico)")
```

```
# Northeast Atlantic EEZs
NEA <- c("Canary Isl. (Spain)", "Spain (mainland; Med and Gulf of Cadiz)", "Spain (Northwes
         "Spain (mainland, Med and Gulf of Cadiz)",
         "France (Atlantic Coast)", "Portugal (mainland)", "Azores Isl. (Portugal)")

# Eastern Central Atlantic EEZs
ECA <- c("Madeira Isl. (Portugal)", "Cape Verde", "Equatorial Guinea", "Benin",
          "Sao Tome & Principe ", "Sao Tome & Principe", "Morocco (Central)", "Ghana",
          "Côte d'Ivoire", "CÃ´te d'Ivoire", "Guinea", "Liberia", "Sierra Leone", "Nigeria"
          "Guinea-Bissau", "Senegal", "Cameroon", "Gambia", "Mauritania", "Morocco (South)"
          "Gabon", "Congo; R. of", "Congo, R. of", "Congo (ex-Zaire)")

# Southeastern Atlantic EEZs
SEA <- c("Angola", "Namibia", "South Africa (Atlantic and Cape)",
           "Ascension Isl. (UK)", "Saint Helena (UK)", "Tristan da Cunha Isl. (UK)")

# Southwest Atlantic EEZs
SWA <- c("Trindade & Martim Vaz Isl. (Brazil)", "Uruguay")

# Mediterranean EEZs
Med <- c("Italy (mainland)", "Libya", "Malta", "Syria", "Sicily (Italy)","Sardinia (Italy)"
         "Balearic Islands (Spain)", "Cyprus (North)", "Cyprus (South)",
         "Gaza Strip", "Greece (without Crete)", "Crete (Greece)", "Lebanon",
         "Turkey (Mediterranean Sea)", "Egypt (Mediterranean)", "Israel (Mediterranean)", "
         "Algeria", "Albania", "Croatia", "Montenegro", "Morocco (Mediterranean)",
         "France (Mediterranean)", "Corsica (France)")

# add variable to specify region
d$reg <- "other oceans"
d$reg[which(d$area_name %in% WCA)] <- "Western Central Atlantic EEZs"
d$reg[which(d$area_name %in% Bra)] <- "Brazil EEZ"
d$reg[which(d$area_name %in% Can)] <- "Canadian EEZ"
d$reg[which(d$area_name %in% USA)] <- "United States EEZ"
d$reg[which(d$area_name %in% NEA)] <- "Northeast Atlantic EEZs"
d$reg[which(d$area_name %in% ECA)] <- "Eastern Central Atlantic EEZs"
d$reg[which(d$area_name %in% SEA)] <- "Southeast Atlantic EEZs"
d$reg[which(d$area_name %in% SWA)] <- "Southwest Atlantic EEZs"
d$reg[which(d$area_name %in% Med)] <- "Mediterranean Sea EEZs"
```

Now we will check that we have classified the regions correctly.

```
# check region designation
othlis <- unique(d$area_name[which(d$reg == "other oceans")])
print("EEZs of all other oceans: ")
```

```
[1] "EEZs of all other oceans: "
```

```
othlis
```

```
 [1] "Australia"                            "China"
 [3] "Taiwan"                               "Mayotte (France)"
 [5] "French Polynesia"                     "Hong Kong (China)"
 [7] "Andaman & Nicobar Isl. (India)"       "Indonesia (Eastern)"
 [9] "Japan (main islands)"                 "Japan (Daito Islands)"
[11] "Korea (South)"                        "Hawaii Northwest Islands (USA)"
[13] "Nauru"                                "Niue (New Zealand)"
[15] "Northern Marianas (USA)"              "Micronesia (Federated States of)"
[17] "Marshall Isl."                        "Palau"
[19] "Pakistan"                             "Papua New Guinea"
[21] "Philippines"                          "Seychelles"
[23] "Viet Nam"                             "Tokelau (New Zealand)"
[25] "Tonga"                                "Hawaii Main Islands (USA)"
[27] "USA (West Coast)"                     "Wake Isl. (USA)"
[29] "Wallis & Futuna Isl. (France)"        "Samoa"
[31] "Guatemala (Pacific)"                  "Kiribati (Gilbert Islands)"
[33] "Mexico (Pacific)"                     "Japan (Ogasawara Islands)"
[35] "Korea (North, Yellow Sea)"            "Korea (North, Sea of Japan)"
[37] "Mauritius"                            "Solomon Isl."
[39] "Christmas Isl. (Australia)"           "Johnston Atoll (USA)"
[41] "Timor Leste"                          "Palmyra Atoll & Kingman Reef (USA)"
[43] "Jarvis Isl. (USA)"                    "Howland & Baker Isl. (USA)"
[45] "Indonesia (Central)"                  "Indonesia (Indian Ocean)"
[47] "Kiribati (Line Islands)"              "Kiribati (Phoenix Islands)"
[49] "New Caledonia (France)"               "Vanuatu"
[51] "Lord Howe Isl. (Australia)"           "Myanmar"
[53] "Ecuador (mainland)"                   "El Salvador"
[55] "Fiji"                                 "Malaysia (Peninsula West)"
[57] "Norfolk Isl. (Australia)"             "Peru"
[59] "Pitcairn (UK)"                        "Colombia (Pacific)"
[61] "Costa Rica (Pacific)"                 "Nicaragua (Pacific)"
```

```
 [63] "Panama (Pacific)"                    "Thailand (Andaman Sea)"
 [65] "Thailand (Gulf of Thailand)"         "Cook Islands"
 [67] "Kermadec Isl. (New Zealand)"         "Tuvalu"
 [69] "Honduras (Pacific)"                  "American Samoa"
 [71] "Galapagos Isl. (Ecuador)"            "New Zealand"
 [73] "Brunei Darussalam"                   "Malaysia (Sabah)"
 [75] "Malaysia (Sarawak)"                  "Guam (USA)"
 [77] "Clipperton Isl. (France)"            "South Africa (Indian Ocean Coast)"
 [79] "Singapore"                           "Easter Isl. (Chile)"
 [81] "Madagascar"                          "Réunion (France)"
 [83] "Chagos Archipelago (UK)"             "Comoros Isl."
 [85] "Mozambique Channel Isl. (France)"    "Kenya"
 [87] "Mozambique"                          "Somalia"
 [89] "Tanzania"                            "Glorieuse Islands (France)"
 [91] "Yemen (Socotra)"                     "Oman"
 [93] "St Paul & Amsterdam Isl. (France)"  "Iran (Sea of Oman)"
 [95] "Desventuradas Isl. (Chile)"          "India (mainland)"
 [97] "Maldives"                            "Tromelin Isl. (France)"
 [99] "Cambodia"                            "Sri Lanka"
[101] "Malaysia (Peninsula East)"           "Bangladesh"
```

```r
print("U.S. EEZs outside the Atlantic Ocean: ")
```

```
[1] "U.S. EEZs outside the Atlantic Ocean: "
```

```r
othlis[grep("USA", othlis)]
```

```
 [1] "Hawaii Northwest Islands (USA)"     "Northern Marianas (USA)"
 [3] "Hawaii Main Islands (USA)"          "USA (West Coast)"
 [5] "Wake Isl. (USA)"                    "Johnston Atoll (USA)"
 [7] "Palmyra Atoll & Kingman Reef (USA)" "Jarvis Isl. (USA)"
 [9] "Howland & Baker Isl. (USA)"         "Guam (USA)"
```

```r
print("French EEZs outside the Atlantic Ocean: ")
```

```
[1] "French EEZs outside the Atlantic Ocean: "
```

```r
othlis[grep("France", othlis)]
```

```
[1] "Mayotte (France)"                  "Wallis & Futuna Isl. (France)"
[3] "New Caledonia (France)"            "Clipperton Isl. (France)"
[5] "Réunion (France)"                  "Mozambique Channel Isl. (France)"
[7] "Glorieuse Islands (France)"        "St Paul & Amsterdam Isl. (France)"
[9] "Tromelin Isl. (France)"
```

```r
unique(d$area_name[which(d$reg == "Western Central Atlantic EEZs")])   # Western Central EE
```

```
 [1] "Bahamas"
 [2] "Barbados"
 [3] "Bermuda (UK)"
 [4] "Cayman Isl. (UK)"
 [5] "Dominica"
 [6] "Dominican Republic"
 [7] "Grenada"
 [8] "Guadeloupe  (France)"
 [9] "Haiti"
[10] "Jamaica"
[11] "Martinique (France)"
[12] "Montserrat (UK)"
[13] "Puerto Rico (USA)"
[14] "Saint Kitts & Nevis"
[15] "Saint Lucia"
[16] "Saint Vincent & the Grenadines"
[17] "Trinidad & Tobago"
[18] "US Virgin Isl."
[19] "St Barthelemy (France)"
[20] "St Martin (France)"
[21] "Curaçao (Netherlands)"
[22] "Bonaire (Netherlands)"
[23] "Saba and Sint Eustatius (Netherlands)"
[24] "Sint Maarten (Netherlands)"
[25] "Honduras (Caribbean)"
[26] "Colombia (Caribbean)"
[27] "Mexico (Atlantic)"
[28] "Turks & Caicos Isl. (UK)"
[29] "Costa Rica (Caribbean)"
```

```
[30] "Panama (Caribbean)"
[31] "Antigua & Barbuda"
[32] "Belize"
[33] "British Virgin Isl. (UK)"
[34] "Cuba"
[35] "French Guiana"
[36] "Guyana"
[37] "Anguilla (UK)"
[38] "Suriname"
[39] "Venezuela"
[40] "Guatemala (Caribbean)"
[41] "Aruba (Netherlands)"
[42] "Nicaragua (Caribbean)"
```

```
unique(d$area_name[which(d$reg == "Brazil EEZ")])   # Brazil EEZs:
```

```
[1] "Brazil (mainland)"
[2] "Fernando de Noronha (Brazil)"
[3] "St Paul and St. Peter Archipelago (Brazil)"
```

```
unique(d$area_name[which(d$reg == "Canadian EEZ")])   # Canadian EEZs:
```

```
[1] "Canada (East Coast)"              "Saint Pierre & Miquelon (France)"
```

```
unique(d$area_name[which(d$reg == "United States EEZ")])  # United States EEZs:
```

```
[1] "USA (East Coast)"     "USA (Gulf of Mexico)"
```

```
unique(d$area_name[which(d$reg == "Northeast Atlantic EEZs")])   # Northeast Atlantic EEZs:
```

```
[1] "Azores Isl. (Portugal)"
[2] "Spain (mainland, Med and Gulf of Cadiz)"
[3] "Canary Isl. (Spain)"
[4] "Portugal (mainland)"
[5] "France (Atlantic Coast)"
[6] "Spain (Northwest)"
```

```r
unique(d$area_name[which(d$reg == "Eastern Central Atlantic EEZs")])   # Eastern Central At
```

```
 [1] "Cape Verde"                "Equatorial Guinea"
 [3] "Sao Tome & Principe"       "Madeira Isl. (Portugal)"
 [5] "Togo"                      "Ghana"
 [7] "Guinea"                    "Liberia"
 [9] "Guinea-Bissau"             "Senegal"
[11] "Sierra Leone"              "Morocco (Central)"
[13] "Morocco (South)"           "Côte d'Ivoire"
[15] "Gabon"                     "Benin"
[17] "Gambia"                    "Mauritania"
[19] "Nigeria"                   "Cameroon"
[21] "Congo, R. of"              "Congo (ex-Zaire)"
```

```r
unique(d$area_name[which(d$reg == "Southeast Atlantic EEZs")])   # Southeast Atlantic EEZs:
```

```
[1] "Ascension Isl. (UK)"                "South Africa (Atlantic and Cape)"
[3] "Saint Helena (UK)"                  "Angola"
[5] "Namibia"                            "Tristan da Cunha Isl. (UK)"
```

```r
unique(d$area_name[which(d$reg == "Southwest Atlantic EEZs")])   # Southwest Atlantic EEZs:
```

```
[1] "Trindade & Martim Vaz Isl. (Brazil)" "Uruguay"
```

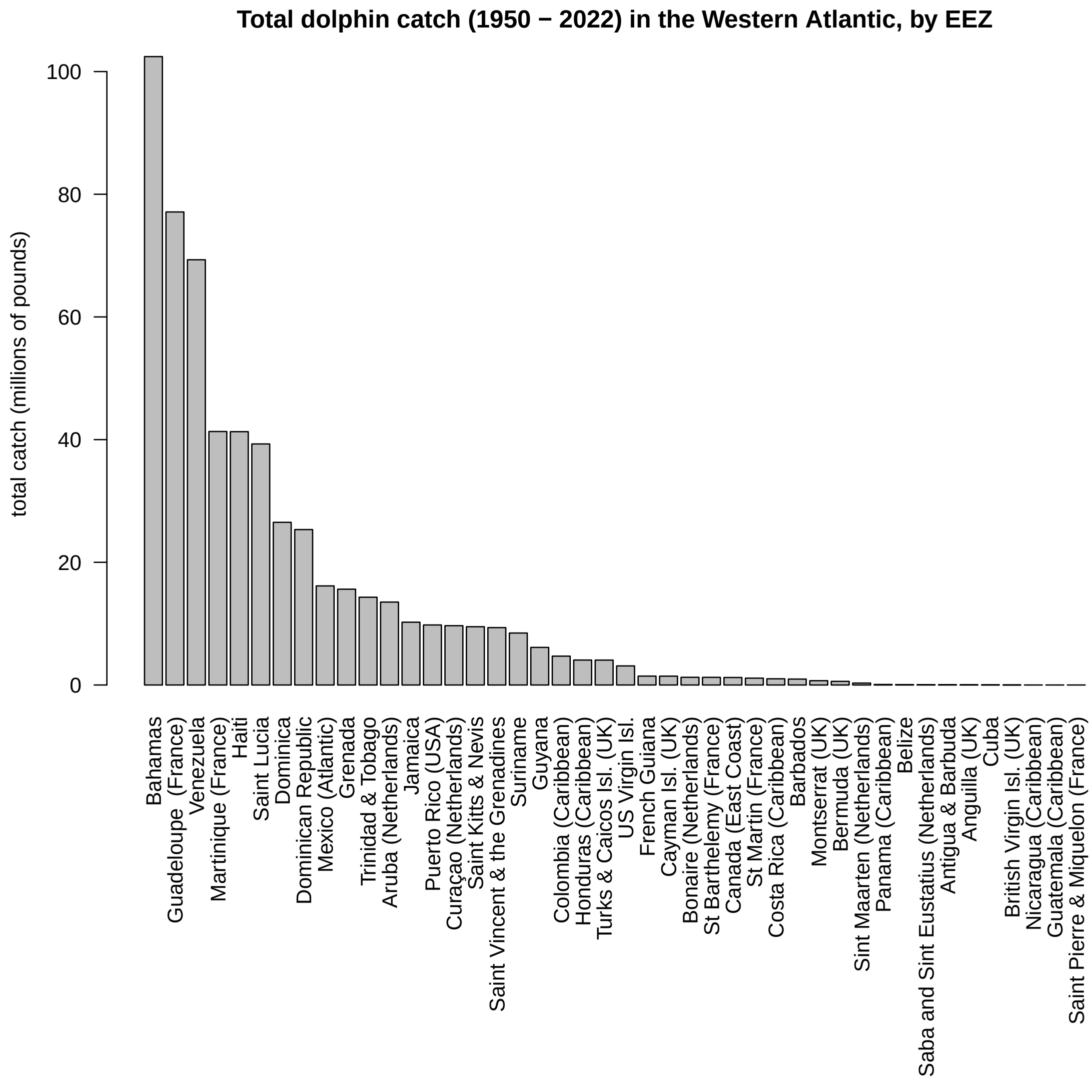
```r
unique(d$area_name[which(d$reg == "Mediterranean Sea EEZs")])   # Mediterranean Sea EEZs:
```

```
 [1] "Italy (mainland)"          "Libya"
 [3] "Malta"                     "Syria"
 [5] "Sicily (Italy)"            "Sardinia (Italy)"
 [7] "Balearic Islands (Spain)" "Tunisia"
 [9] "Algeria"                   "Morocco (Mediterranean)"
[11] "France (Mediterranean)"    "Corsica (France)"
[13] "Croatia"                   "Lebanon"
[15] "Albania"                   "Greece (without Crete)"
[17] "Crete (Greece)"            "Cyprus (South)"
```

12.4 Summarize catch by regional EEZs

Now we summarize the catch by region by grouping the catch across the EEZs by region. We save these data for later analysis, and move on to analyzing only the Western Central and Northwest Atlantic jurisdictions.

```
# summarize EEZ catch by region
tab <- tapply(d$lbs, list(d$year, d$reg), sum, na.rm = T)
write.csv(data.frame(tab), file = "data/outputs/intl_catches_allAtlantic.csv", row.names =

dwest <- d[which(d$reg == "Canadian EEZ" | d$reg == "Western Central Atlantic EEZs"), ]
save(dwest, file = "data/outputs/SAU_EEZs_WCA.RData")

tab <- tapply(dwest$lbs, list(dwest$year, dwest$area_name), sum, na.rm = T)

# eight countries are responsible for vast majority of catch with total catches across all
par(mar = c(17, 4, 3, 1))
barplot(sort(colSums(tab, na.rm = T)/10^6, decreasing = T), horiz = F, las = 2,
        main = "Total dolphin catch (1950 - 2022) in the Western Atlantic, by EEZ",
        ylab = "total catch (millions of pounds)")
```



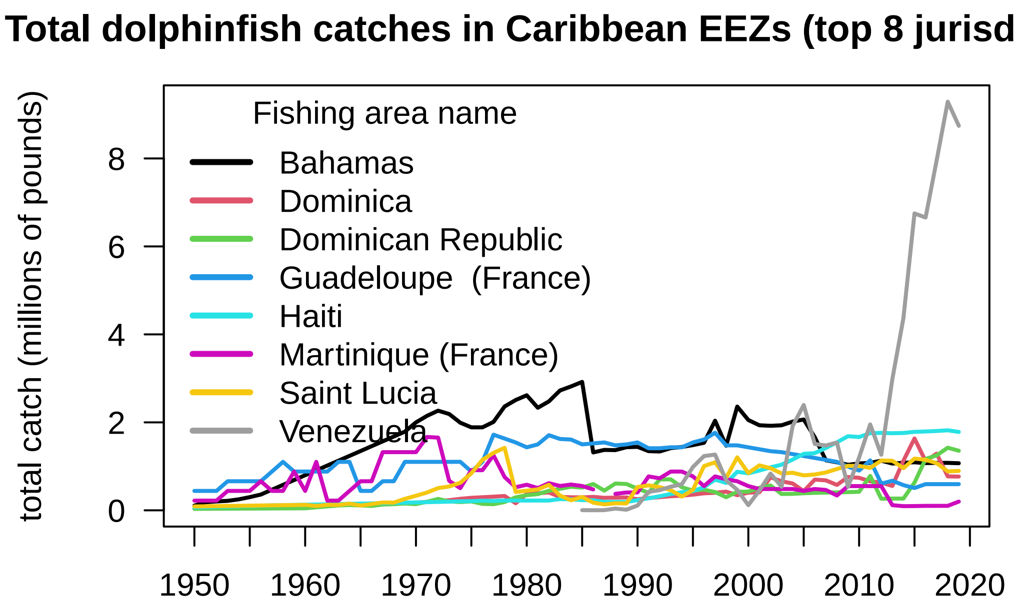
Note that Bermuda is actually in the NCA region and not the Greater Caribbean region. However, its catches constitute 0.1% of the total catch which is essentially negligible, and to avoid complicating the operating model and code unnecessarily we will leave those catches accounted for in the CAR region.

```
round(colSums(tab, na.rm = T)[7] / sum(colSums(tab, na.rm = T), na.rm = T) * 100, 2)
```

```
Bermuda (UK)
```

```
0.1
```

```
lis <- names(sort(colSums(tab, na.rm = T), decreasing = T)[1:8]) # list of top 8 jurisdicti
tab1 <- tab[, colnames(tab) %in% lis]

par(mar = c(3, 5, 3, 1))
matplot(rownames(tab1), tab1/10^6, las = 2, type = "l", lty = 1, lwd = 2, pch = 19, col = 1
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Total dolphinfish catches in Caribbean EEZs (top 8 jurisdictions)")
axis(1, at = seq(1950, 2020, 5)); axis(2, las = 2); box()
legend("topleft", colnames(tab1), col = 1:8, lty = 1, lwd = 3, bty = "n", title = "Fishing
```

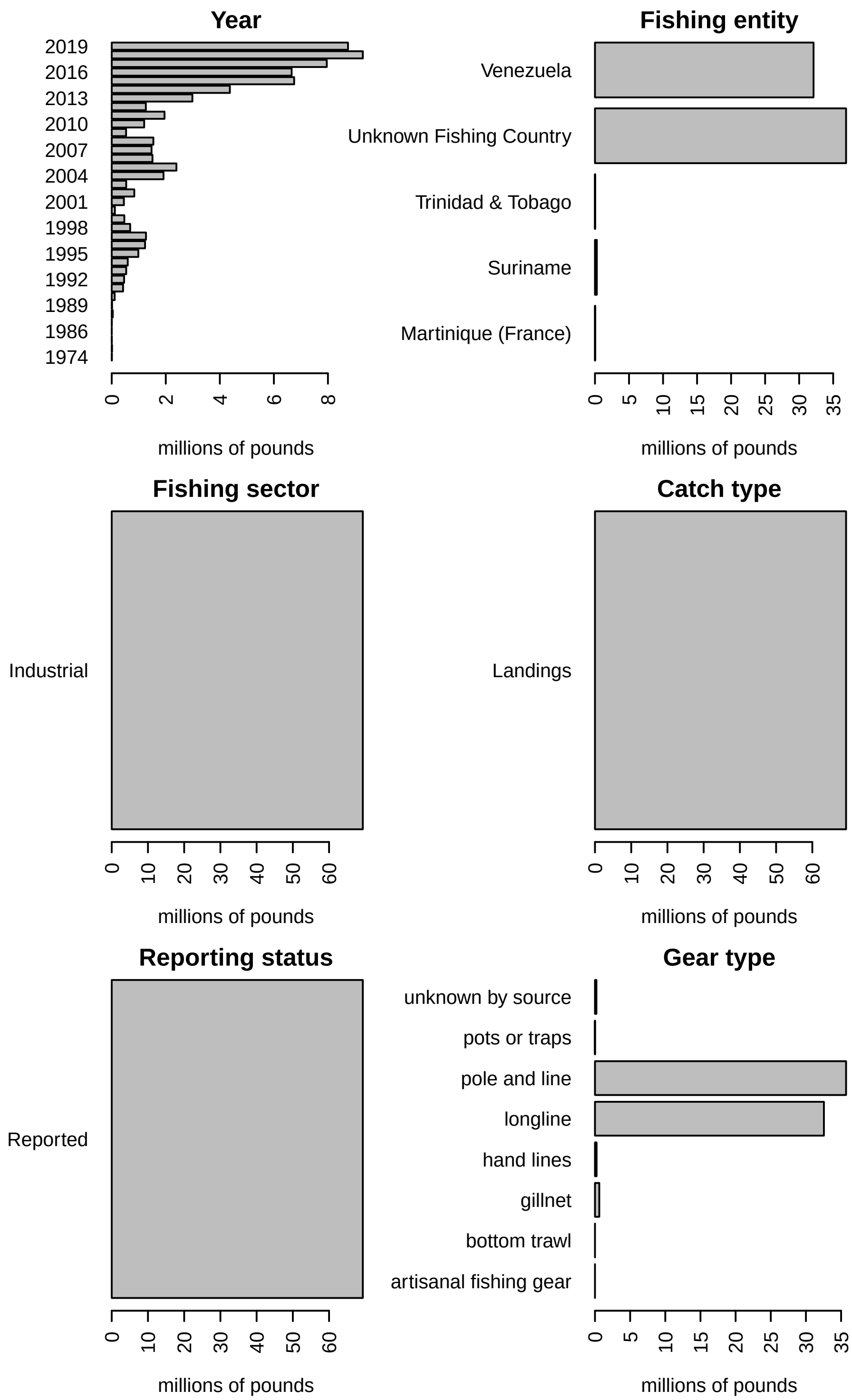


Note the rapid eight-fold increase in catches listed within the Venezuelan EEZ which occurs from 2013 - 2019. This increase is responsible for a rapid increase in estimated international catches within the Western Atlantic.

12.5 Look at Venezuela catches in detail

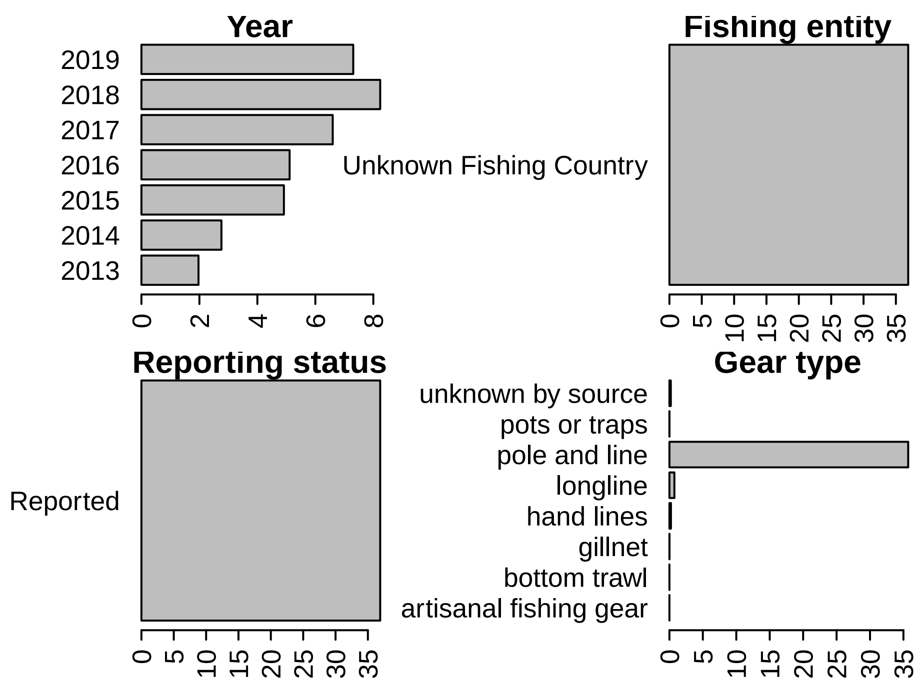
The figures above indicate that the database shows a rapid increase in international catches within the Western Atlantic, driven by a sudden increase in catches reported within Sea Around Us database within the Venezuelan EEZ. We investigate these catches in detail, as it seems odd that a relatively small fleet from a single country would be able to ramp up effort so rapidly. Data can be downloaded on a country-specific basis, the catches by taxon in the waters of Venezuela are accessed here.

```
rm(list = ls())  # clear the workspace

# download the data for waters of Venezuela
ven <- read.csv("data/SAU/SAU EEZ 862 v50-1.csv")

# subset dolphinfish from the data
v <- ven[which(ven$common_name == "Common dolphinfish"), ]

#apply(v[1:13], 2, table)

#convert to pounds
v$pounds <- v$tonnes * 1000 * 2.20462

# look at the characteristics of the data
par(mar = c(5, 10, 1, 1), mfrow = c(3, 2), mex = 0.9)
#barplot(tapply(v$pounds/10^6, v$area_name, sum, na.rm = T), las = 2, xlab = "millions of p
#barplot(tapply(v$pounds/10^6, v$area_type, sum, na.rm = T), las = 2, xlab = "millions of p
barplot(tapply(v$pounds/10^6, v$year, sum, na.rm = T), las = 2, xlab = "millions of pounds"
barplot(tapply(v$pounds/10^6, v$fishing_entity, sum, na.rm = T), las = 2, xlab = "millions 
barplot(tapply(v$pounds/10^6, v$fishing_sector, sum, na.rm = T), las = 2, xlab = "millions 
barplot(tapply(v$pounds/10^6, v$catch_type, sum, na.rm = T), las = 2, xlab = "millions of p
barplot(tapply(v$pounds/10^6, v$reporting_status, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(v$pounds/10^6, v$gear_type, sum, na.rm = T), las = 2, xlab = "millions of po
```


Year
2019
2016
2013
2010
2007
2004
2001
1998
1995
1992
1989
1986
1974
0 2 4 6 8
millions of pounds
Fishing entity
Venezuela
Unknown Fishing Country
Trinidad & Tobago
Suriname
Martinique (France)
0 5 10 15 20 25 30 35
millions of pounds
Fishing sector
Industrial
0 10 20 30 40 50 60
millions of pounds
Catch type
Landings
0 10 20 30 40 50 60
millions of pounds
Reporting status
Reported
0 10 20 30 40 50 60
millions of pounds
Gear type
unknown by source
pots or traps
pole and line
longline
hand lines
gillnet
bottom trawl
artisanal fishing gear
0 5 10 15 20 25 30 35
millions of pounds

Note that a large portion of the catch in the database is listed as coming from an "Unknown Fishing Country." We extract these catches and further investigate the characteristics.

```
# extract the Unknown Fishing Country catch
v1 <- v[which(v$fishing_entity == "Unknown Fishing Country"), ]
#apply(v1[1:15], 2, table)

# look at characteristics
par(mar = c(3, 10, 1, 5), mfrow = c(2, 2), mex = 0.6)
barplot(tapply(v1$pounds/10^6, v1$year, sum, na.rm = T), las = 2, xlab = "millions of pound
barplot(tapply(v1$pounds/10^6, v1$fishing_entity, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(v1$pounds/10^6, v1$reporting_status, sum, na.rm = T), las = 2, xlab = "milli
barplot(tapply(v1$pounds/10^6, v1$gear_type, sum, na.rm = T), las = 2, xlab = "millions of
```

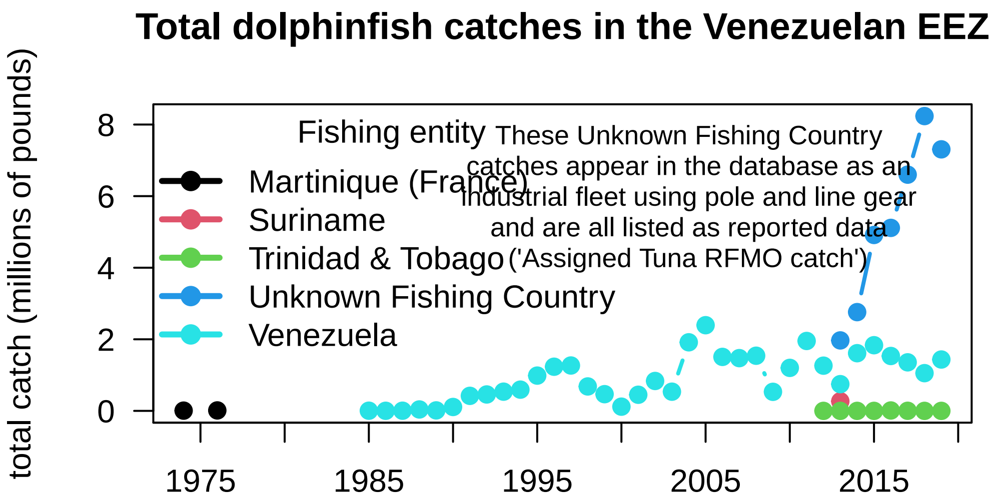


Here is another look at the catches within the venezuelan EEZ, separated by fishing entity.

```
# summarize Venezuela EEZ catches by fishing entity
tab <- tapply(v$pounds, list(v$year, v$fishing_entity), sum, na.rm = T)
head(tab)
```

```
     Martinique (France) Suriname Trinidad & Tobago Unknown Fishing Country
1974            7320.429       NA                NA                      NA
1976           12814.641       NA                NA                      NA
1985                  NA       NA                NA                      NA
1986                  NA       NA                NA                      NA
1987                  NA       NA                NA                      NA
1988                  NA       NA                NA                      NA
     Venezuela
1974        NA
1976        NA
1985  4699.563
1986  2229.484
1987  4585.610
1988 39960.149
```

```
matplot(rownames(tab), tab/10^6, las = 2, type = "b", lty = 1, lwd = 2, pch = 19, col = 1:5
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Total dolphinfish catches in the Venezuelan EEZ")
axis(1, at = seq(1950, 2020, 5)); axis(2, las = 2); box()
legend("topleft", colnames(tab), col = 1:5, pch = 19, lty = 1, lwd = 3, bty = "n", title =

text(2004, 6, cex = 0.8,
     "These Unknown Fishing Country\ncatches appear in the database as an\nindustrial fleet
```



The very substantial increase in international Western Atlantic dolphin catches can be attributed solely to this Unknown Fishing Country catches which appear in the database beginning in 2013. The documentation of the catch reconstruction can be found in the Sea Around Us documentation files (see Mendoza et al. 2015). We spoke to Jeremy Mendoza and other experts who were involved in the reconstruction of domestic catches in Venezuela, and they noted that the "Assigned tuna RFMO catch" actually emerge from spatially reshuffled landing reports to ICCAT, according to the methods of Coulter et al. (2020). In this work, the Sea Around Us team developed a global spatial reallocation of landings of tunas and associated species that were reported to ICCAT. In short, this work uses the fraction of reported spatially disaggregated data to reallocate all data reported to ICCAT. This step is necessary because there are not many countries that report spatial data to ICCAT in the region, and this method may lead to a significant reallocation of landings from the Caribbean to the Venezuelan EEZ, where the local pole and line fishery is significant (J. Mendoza, personal communication). Thus, it is likely that landings from other EEZs, that were included in the reconstructed catch data layer, could have been reassigned to the Venezuelan EEZ.

Because there may be some overlap between the reconstructed data used in the national EEZ estimates, and the ICCAT reporting that was reassigned, there could potentially be a "double counting" of these catch estimates in the Sea Around Us database. We verified with Sea Around Us database managers that their method led to an error resulting in an overestimate of the catch, and they advised us to remove these Unknown Fishing Country catches as they are difficult to trace back to the data source.

12.6 Compile catch in CAR and NED jurisdictional waters by discards and reporting type

Now that we have separated the EEZ catch by region, we will summarize it according to the areas used in the dolphin management strategy evaluation operating model. We will separate the catch by discards versus landings and reported versus unreported data so that we can use these data in various sensitivity analyses.

```
rm(list = ls())  # clear the workspace

# load data
load("data/outputs/SAU_EEZs_WCA.RData")  # SAU EEZ data for NED and NCA saved from earlier
load("data/outputs/erroneous_catch_to_be_removed.RData")   # double-counted catch values to

d <- dwest

# summarize the catch by area, discards/landings and reported/unreported
tapply(d$lbs, list(d$catch_type, d$reporting_status, d$reg), sum, na.rm = T)
```

```
, , Canadian EEZ

         Reported Unreported
Discards       NA         NA
Landings  1220146         NA

, , Western Central Atlantic EEZs

          Reported Unreported
Discards        NA    2179444
Landings 393279187  176215080
```

The catch from the Canadian EEZs is 100% landings, with no unreported data and no discards. In the WCA there are no reported discards, only unreported discards. Thus, we only need to separate out unreported discards and landings for the WCA.

```
# summarize the catch by year, area, discards/landings and reported/unreported

d$year <- factor(d$year, levels = 1950:2019)
dc <- d[which(d$reg == "Canadian EEZ"), ]
dw <- d[which(d$reg == "Western Central Atlantic EEZs"), ]

# verify proper identification of the erroneous Venezuelan Unknown Fishing Country catches
bad <- dw[which(dw$area_name == "Venezuela" & dw$fishing_entity == "Unknown Fishing Country
#dim(bad)
bad_totals <- tapply(bad$lbs, bad$year, sum, na.rm = T)

cbind(tail(bad_totals, 10), tail(removecatch, 10)) # the numbers match exactly
```

```
        [,1]    [,2]
2010      NA      NA
2011      NA      NA
2012      NA      NA
2013 1968566 1968566
2014 2759603 2759603
2015 4915156 4915156
2016 5112450 5112450
2017 6599737 6599737
2018 8236857 8236857
2019 7307245 7307245
```

```
# now remove those bad catches from the database prior to summarizing
#dim(dw)
dw <- dw[-which(dw$area_name == "Venezuela" & dw$fishing_entity == "Unknown Fishing Country
#dim(dw)

CAN <- tapply(dc$lbs, list(dc$year, dc$catch_type, dc$reporting_status), sum, na.rm = T)
tail(CAN)
```

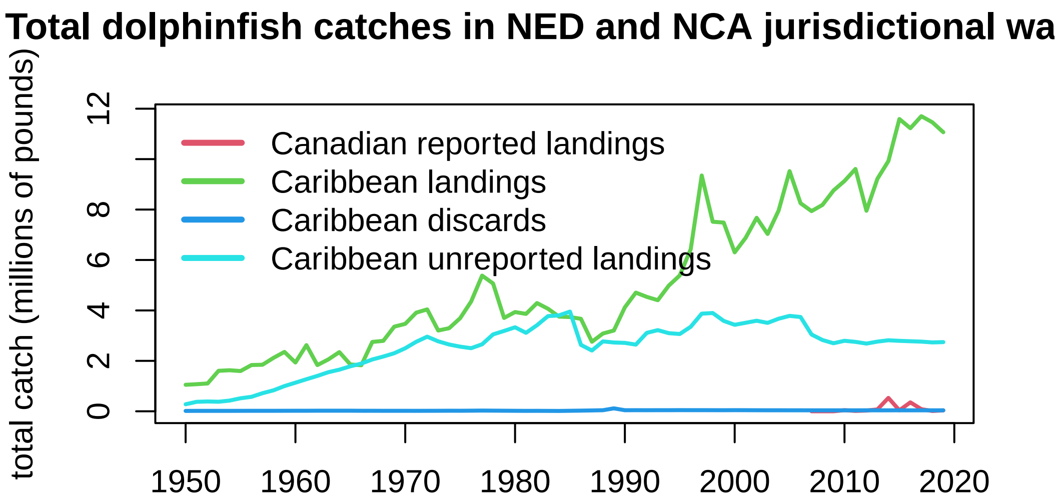
```
, , Reported

      Landings
2014 531022.55
2015  38945.76
2016 354370.65
```

```
2017  83776.88
2018  12588.88
2019  36112.19
```

```
tab_dw <- tapply(dw$lbs, list(dw$year, dw$catch_type, dw$reporting_status), sum, na.rm = T)
tail(tab_dw)
```

```
, , Reported

     Discards Landings
2014       NA  9923818
2015       NA 11590469
2016       NA 11225710
2017       NA 11702729
2018       NA 11460758
2019       NA 11066090

, , Unreported

     Discards Landings
2014 38062.63  2816498
2015 39134.61  2794662
2016 38973.17  2778493
2017 38728.81  2762926
2018 38769.93  2732351
2019 38655.51  2743182
```

```
# combine CAR reported landings, unreported discards, unreported landings
CAR <- data.frame(cbind(tab_dw[, 2, 1], tab_dw[, , 2]))
names(CAR) <- c("landings", "discards", "unreported")

matplot(rownames(CAR), cbind(CAN, CAR)/10^6, type = "l", lwd = 2, lty = 1, col = 2:5,
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Total dolphinfish catches in NED and NCA jurisdictional waters")
legend("topleft", col = 2:5, c("Canadian reported landings", "Caribbean landings",
                    "Caribbean discards", "Caribbean unreported landings"), lty = 1, lwd =
```




```
round(sum(CAR$discards, na.rm = T) / (sum(CAR$discards, na.rm = T) + sum(CAR$landings, na.r
```

```
[1] 0.61
```

Note that the Caribbean unreported discards in the database are negligible (< 1%) and we exclude them from further analysis.

12.7 Distribute the territorial international catches across quarters of the year

The international catches are only reported as annual catches and do not have any month or season associated with them. However, the MSE operating model requires seasonal catches. Lacking other information on the seasonality of these fleets, we will assume that the fleet distribution of the U.S. pelagic longline is representative of the seasonality of the jurisdictional fleets. Now we parse the annual catches according to the seasonality of the U.S. fleet.

The pelagic longline logbook data is only available going back to 1997, but there are Caribbean catches extending back to the 1950s. To interpolate the seasonality for

1986 - 1996, we take the average seasonality of the oldest 10 years of logbook data (1997 - 2006) and use those averages for the unknown years.

At the time of analysis, the SAU database only contained catches to 2019, so we interpolated the 2019 values to cover years 2020 - 2022.

```
#par(mar = c(5, 10, 3, 1))
#barplot(tapply(dw$lbs, dw$gear_type, sum, na.rm = T), horiz = T, las = 2)
#barplot(tapply(dc$lbs, dc$gear_type, sum, na.rm = T), horiz = T, las = 2)

# read in the percentage of the catch by area and year-quarter combinations from the U.S. P
load("data/outputs/per_PLLcatch_by_area_yearquarter.RData")

names(percatch[1, 1, ]) # matrix 2 == Caribbean, matrix 7 == NED (Atlantic Northwest)
```

```
[1] "NCA"  "CAR"  "FLK"  "NCFL" "NNC"  "VBM"  "NED"
```

```
CAN <- CAN[which(rownames(CAN) >= 1997)]  # PLL data are only available from 1997 on
CAN <- c(CAN, CAN[length(CAN)], CAN[length(CAN)], CAN[length(CAN)])  # interpolate 2019 cat

CANqrt <- percatch[, , 7]   # create matrix for final catch data
for (i in 1:ncol(percatch))  {
  CANqrt[, i] <- percatch[, i, 7] * CAN[i]  }

percatch_car <- percatch[, , 2] # create matrix for Caribbean percentages
int <- rowMeans(percatch_car[, 1:10]) # calculate the average seasonality using the oldest
percatch_car <- cbind(matrix(rep(int, length(1986:1996)), nrow = 4), percatch_car)  # attac
colnames(percatch_car) <- 1986:2022   # full data frame for Caribbean seasonality 1986 - 20

CAR <- CAR[which(rownames(CAR) >= 1986), ]  # only need Caribbean data from 1986 on
CAR <- rbind(CAR, CAR[nrow(CAR), ], CAR[nrow(CAR), ], CAR[nrow(CAR), ])  # interpolate 2019

#dim(CAR)
#dim(percatch_car)

CARrep <- percatch_car   # create matrices for final catch data
CARunr <- percatch_car

for (i in 1:ncol(percatch_car))  {
  CARrep[, i] <- percatch_car[, i] * CAR[i, 1]
```

```
  CARunr[, i] <- percatch_car[, i] * CAR[i, 3]
}

# ensure sums are equal after parsing
sum(CARrep); sum(CAR$landings)
```

```
[1] 286220280
```

```
[1] 286220280
```

```
sum(CARunr); sum(CAR$unreported)
```

```
[1] 112754888
```

```
[1] 112754888
```

```
sum(CAN, na.rm = T); sum(CANqrt, na.rm = T)
```

```
[1] 1328482
```

```
[1] 1328482
```

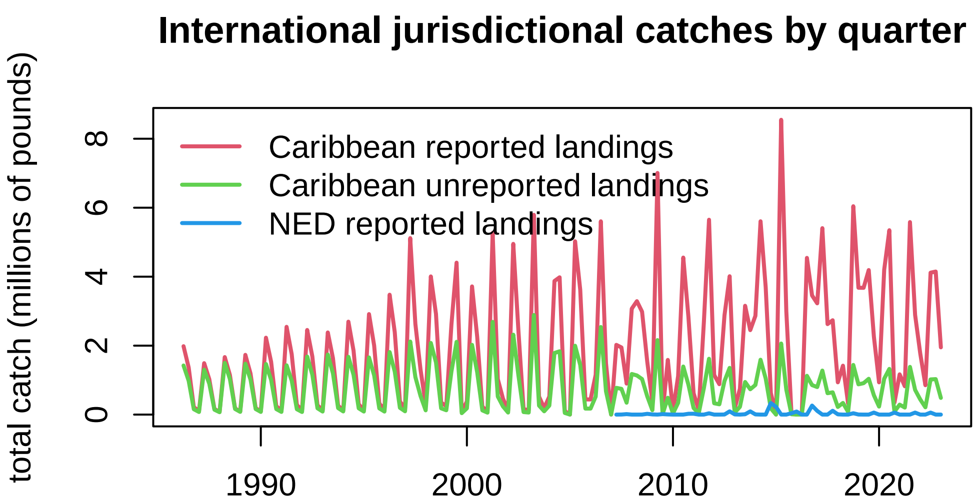
The Canada jurisdictional data need to be combined with the NED high seas data calculated in the previous section, because both of these are a component of the international catches in the NED area.

```
load("data/outputs/NED_year_quarter.RData")

NEDtot <- NED + CANqrt
# ensure sums are correct
sum(NED, na.rm = T) + sum(CANqrt, na.rm = T); sum(NEDtot, na.rm = T)
```

```
[1] 1808855
```

```
[1] 1808855
```

```
NEDtot <- NEDtot[, !is.na(colSums(NEDtot))] # remove NA columns from CAN data

# plot the parsed out data by year-quarter
yrs <- sort(rep(as.numeric(colnames(CARrep)), 4))
qrt <- rep(1:4, length(yrs)/4)
yrs1 <- sort(rep(as.numeric(colnames(NEDtot)), 4))
qrt1 <- rep(1:4, length(yrs1)/4)

matplot(yrs + qrt/4, cbind(matrix(CARrep), matrix(CARunr))/10^6, lty = 1, lwd = 2,
        col = 2:3, type = "l", main = "International jurisdictional catches by quarter",
        xlab = "", ylab = "total catch (millions of pounds)")
lines(yrs1 + qrt1/4, as.numeric(matrix(NEDtot))/10^6, lwd = 2, col = 4)
legend("topleft", c("Caribbean reported landings", "Caribbean unreported landings", "NED re
       lwd = 2, col = 2:4, bty = "n")
```



Finally, we format the data in the format for input into the operating model.

```
# format for operating model - Caribbean
flt <- rep("Intl", length(yrs))
are <- rep("CAR", length(yrs))
fin1 <- data.frame(cbind(yrs, qrt, flt, are, matrix(CARrep)))
```

```
flt <- rep("UnRep", length(yrs))
fin2 <- data.frame(cbind(yrs, qrt, flt, are, matrix(CARunr)))

findat <- data.frame(rbind(fin1, fin2), stringsAsFactors = FALSE)
names(findat) <- c("Year", "Quarter", "Fleet", "Area", "Catch_lbs")
findat$Catch_lbs <- as.numeric(findat$Catch_lbs)
head(findat)
```

```
  Year Quarter Fleet Area Catch_lbs
1 1986       1  Intl  CAR 1983610.3
2 1986       2  Intl  CAR 1363432.3
3 1986       3  Intl  CAR  216194.7
4 1986       4  Intl  CAR  104564.4
5 1987       1  Intl  CAR 1491251.1
6 1987       2  Intl  CAR 1025009.7
```

```
#plot(findat$Catch_lbs, type = "l")
#plot(findat$Catch_lbs ~ factor(findat$Fleet))
#plot(findat$Catch_lbs ~ factor(findat$Quarter))
#plot(findat$Catch_lbs ~ factor(findat$Year))
#apply(findat, 2, table)

# write the output file
write.csv(findat, file = "data/FINAL_files/CAR_Intl_unrep_TomFormat.csv", row.names = FALSE

# format for operating model - NED
flt <- rep("Intl", length(yrs1))
are <- rep("NED", length(yrs1))
fin <- data.frame(cbind(yrs1, qrt1, flt, are, matrix(NEDtot)), stringsAsFactors = FALSE)
names(fin) <- c("Year", "Quarter", "Fleet", "Area", "Catch_lbs")
fin$Catch_lbs <- as.numeric(fin$Catch_lbs)
head(fin)
```

```
  Year Quarter Fleet Area Catch_lbs
1 2007       1  Intl  NED     0.000
2 2007       2  Intl  NED  1450.366
3 2007       3  Intl  NED 13279.912
4 2007       4  Intl  NED     0.000
5 2008       1  Intl  NED     0.000
6 2008       2  Intl  NED  1312.750
```

```
#apply(fin, 2, table)

# write output file
write.csv(fin, file = "data/FINAL_files/NED_Intl_TomFormat.csv", row.names = FALSE)

#barplot(tapply(fin$Catch_lbs, list(fin$Quarter), sum, na.rm = T), col = rainbow(4))
```

12.8 References

Pauly D., Zeller D., Palomares M.L.D. (Editors), 2020. Sea Around Us Concepts, Design and Data (seaaroundus.org).

13 Data compilation for all fleets in the operating model

In this last step we will compile all of the output data files from the different data sources (NMFS commercial trip tickets, NMFS recreational statistics, Sea Around Us) and check the files for accuracy.

Recall that there are seven defined areas in the operating model, over which the catches are summed. Four of these areas are within the U.S. EEZ and the other three areas are international waters (high seas and jurisdictional waters of other countries).

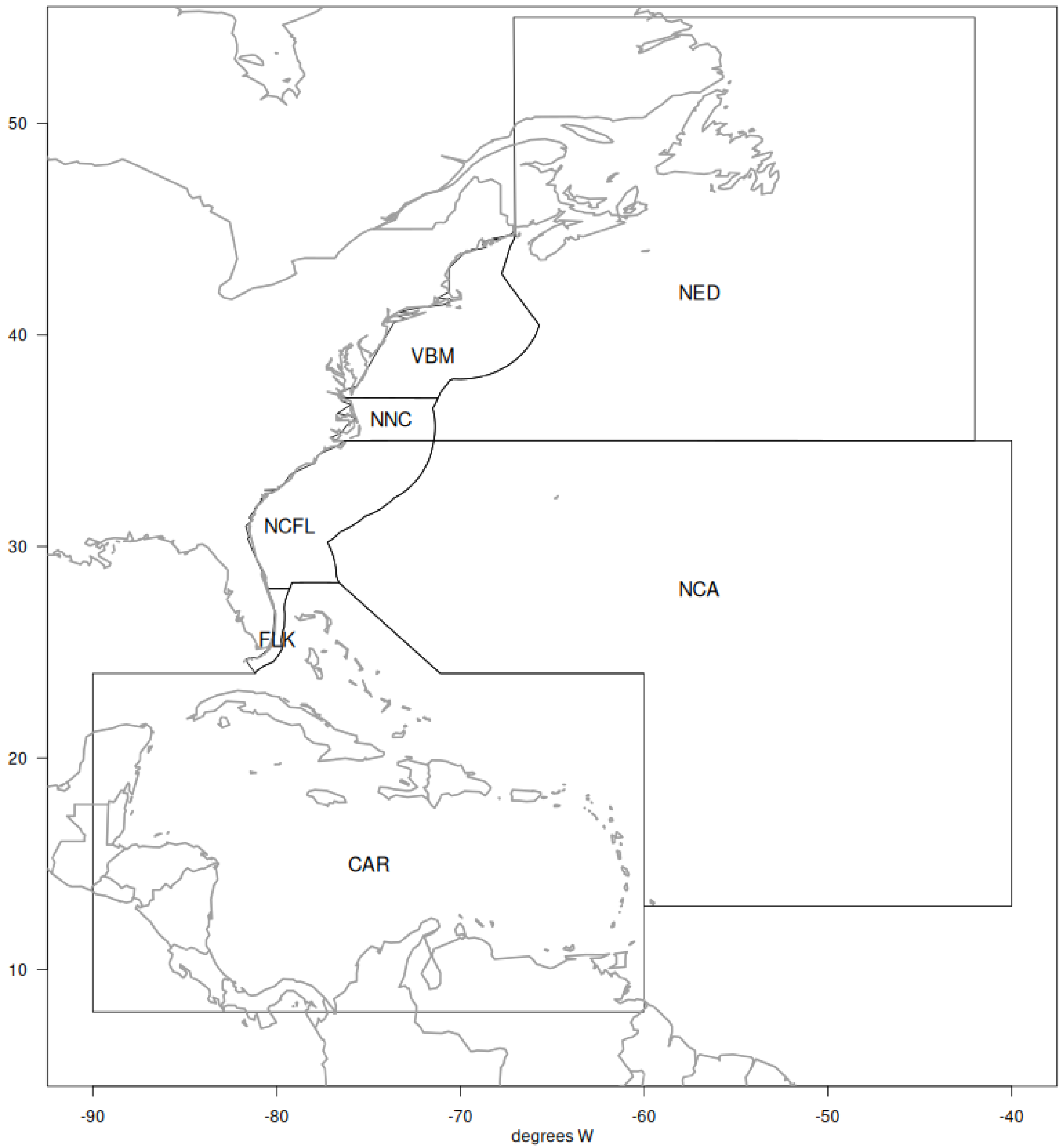


Figure 13.1: Areas defined in the operating model.

13.1 Data upload

First we access all of the final data files from the previous steps.

```
# clear workspace
rm(list = ls())

# view data files in directory
dir("data/FINAL_files/")
```

```
[1] "CAR_Intl_unrep_TomFormat.csv"
[2] "commercial_TomFormat.csv"
[3] "complete_catch_dataset_09052025b.csv"
[4] "NCA_Intl_TomFormat.csv"
[5] "NED_Intl_TomFormat.csv"
[6] "rec_catch_TomFormat.csv"
```

```
intc <- read.csv("data/FINAL_files/CAR_Intl_unrep_TomFormat.csv")
intn <- read.csv("data/FINAL_files/NCA_Intl_TomFormat.csv")
intw <- read.csv("data/FINAL_files/NED_Intl_TomFormat.csv")
com <- read.csv("data/FINAL_files/commercial_TomFormat.csv")
rec <- read.csv("data/FINAL_files/rec_catch_TomFormat.csv")
```

This file contains all international catches and unreported catches for the greater Caribbean region (jurisdictional waters of all countries). Recall that estimated discards were very small (< 1%) and were removed from the summary.

```
apply(intc[1:4], 2, table, useNA = "always")
```

```
$Year

1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001
   8    8    8    8    8    8    8    8    8    8    8    8    8    8    8    8
2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017
   8    8    8    8    8    8    8    8    8    8    8    8    8    8    8    8
2018 2019 2020 2021 2022 <NA>
   8    8    8    8    8    0
```

```
$Quarter

   1    2    3    4 <NA> 
  74   74   74   74    0 

$Fleet

 Intl UnRep  <NA> 
  148   148     0 

$Area

 CAR <NA> 
 296    0 
```

This file contains all international catches for the NCA region (i.e., FAO region 31), also known as the Western Central Atlantic. There are no territorial waters in this area (except for Bermuda, which as discussed previously is summarized with the CAR region) and there are no unreported catches or discards in the database.

```r
apply(intn[1:4], 2, table, useNA = "always")
```

```
$Year

1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 
   4    4    4    4    4    4    4    4    4    4    4    4    4    4    4    4 
2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 <NA> 
   4    4    4    4    4    4    4    4    4    4    0 

$Quarter

   1    2    3    4 <NA> 
  26   26   26   26    0 

$Fleet

Intl <NA> 
 104    0 

$Area
```

```
 NCA <NA> 
 104    0 
```

This file contains all international catches for the NED region (i.e., FAO region 21), also known as the Northwest Atlantic. This includes NED high seas catches, as well as catches from Canadian and French territorial waters which fall in this region. There are no unreported catches or discards for this region in the database.

```
apply(intw[1:4], 2, table, useNA = "always")
```

```
$Year

2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 
   4    4    4    4    4    4    4    4    4    4    4    4    4    4    4    4 
<NA> 
   0 

$Quarter

   1    2    3    4 <NA> 
  16   16   16   16    0 

$Fleet

Intl <NA> 
  64    0 

$Area

 NED <NA> 
  64    0 
```

This file contains all catches for the U.S. commercial fleet, including catches from U.S. EEZs and high seas. There are no unreported catches in this region. Recall that estimated dead discards were very small (< 2%) and were removed from the summary.

```
apply(com[1:4], 2, table, useNA = "always")
```

```
$Year

1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001
  28   28   28   28   28   28   28   28   28   28   28   28   28   28   28   28
2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017
  28   28   28   28   28   28   28   28   28   28   28   28   28   28   28   28
2018 2019 2020 2021 2022 <NA>
  28   28   28   28   28    0

$Quarter

   1    2    3    4 <NA>
 259  259  259  259    0

$Fleet

UScom  <NA>
 1036     0

$Area

 CAR  FLK  NCA NCFL  NED  NNC  VBM <NA>
 148  148  148  148  148  148  148    0
```

This file contains all catches for the U.S. recreational fleet, which are all assumed to occur within the U.S. EEZs. There are no unreported catches in this region.

```r
apply(rec[1:4], 2, table, useNA = "always")
```

```
$Year

1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001
  32   32   32   32   32   32   32   32   32   32   32   32   32   32   32   32
2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017
  32   32   32   32   32   32   32   32   32   32   32   32   32   32   32   32
2018 2019 2020 2021 2022 <NA>
  32   32   32   32   32    0

$Quarter
```

```
   1    2    3    4 <NA> 
 296  296  296  296    0 

$Fleet

Hire  Rec <NA> 
 592  592    0 

$Area

 FLK NCFL  NNC  VBM <NA> 
 296  296  296  296    0 
```

Now we combine the data files and create some figures with the final composite data file.

```
d <- rbind(intc, intn, intw, com, rec)

# how many rows contain zeros
#hist(d$Catch_lbs)
table(d$Catch_lbs == 0)
```

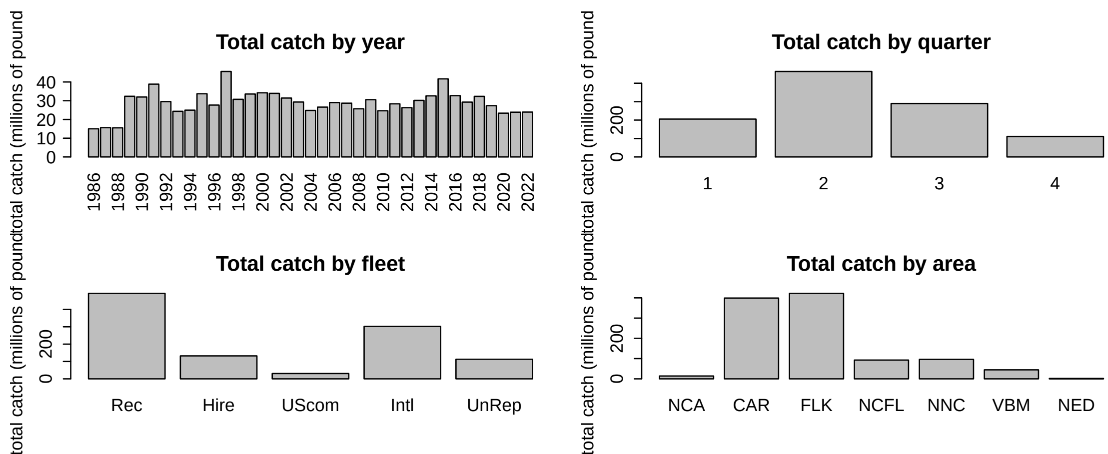
```
FALSE  TRUE 
 2047   637 
```

```
# resort levels geographically
d$Area <- factor(d$Area, levels = c("NCA", "CAR", "FLK", "NCFL", "NNC", "VBM", "NED"))
d$Fleet <- factor(d$Fleet, levels = c("Rec", "Hire", "UScom", "Intl", "UnRep"))
```

13.2 Visualizing the final data set

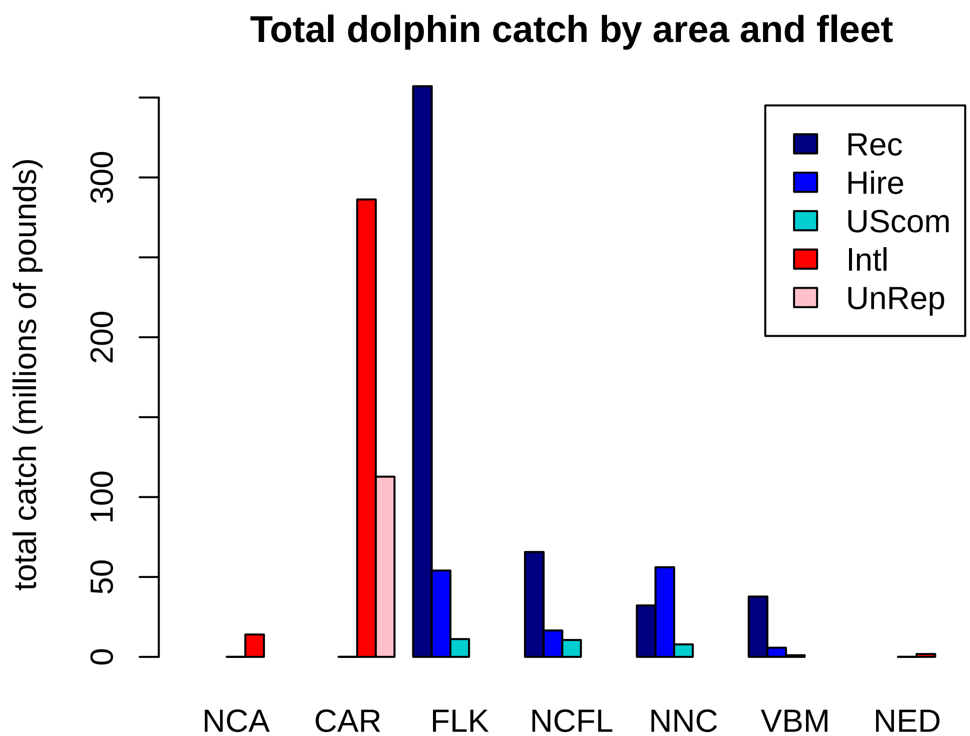
We plot the final data set in a number of different ways to detect any errors that may have occurred during processing.

```
# plot the total catches by each factor
par(mfrow = c(2, 2), mex = 0.7)
barplot(tapply(d$Catch_lbs, d$Year, sum, na.rm = T)/10^6, las = 2,
        main = "Total catch by year", ylab = "total catch (millions of pounds)")
barplot(tapply(d$Catch_lbs, d$Quarter, sum, na.rm = T)/10^6,
        ylab = "total catch (millions of pounds)", main = "Total catch by quarter")
barplot(tapply(d$Catch_lbs, d$Fleet, sum, na.rm = T)/10^6,
        main = "Total catch by fleet", ylab = "total catch (millions of pounds)")
barplot(tapply(d$Catch_lbs, d$Area, sum, na.rm = T)/10^6,
        main = "Total catch by area", ylab = "total catch (millions of pounds)")
```



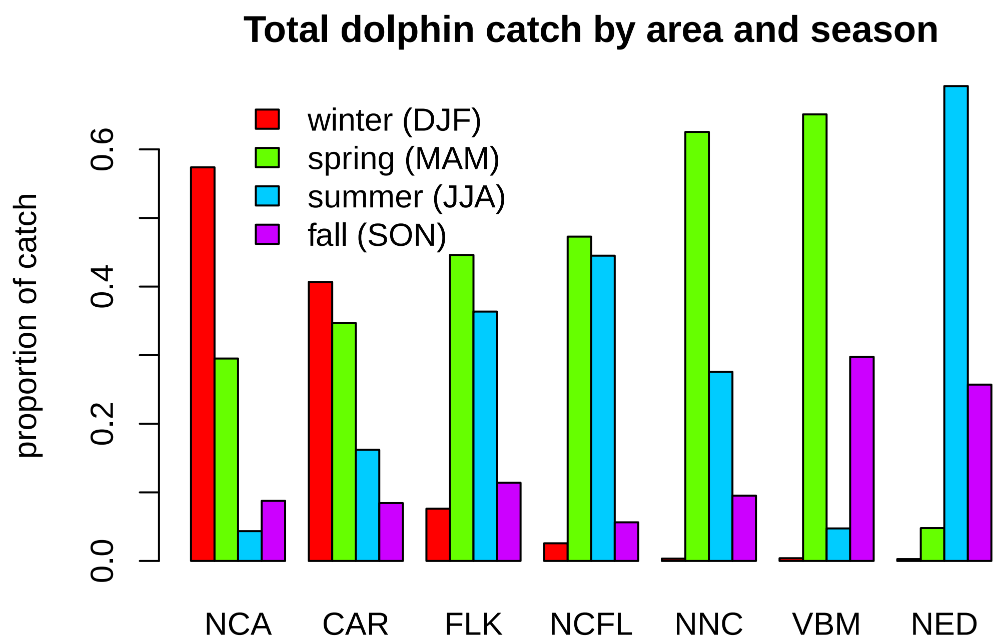
The total catch in the regions analyzed is highly variable but relatively stable over time with a slight decrease in recent years. Most of the catch is caught in quarter 2 (March - May). The largest sector is the U.S. private recreational sector, followed by international fleets (all types), the for-hire fleet and then the U.S. commercial fleet. Most of the fishing activity takes place in the Florida Keys and the Greater Caribbean, with other regions contributing much less.

```
par(mfrow = c(1, 1), mar = c(2, 5, 3, 1))
tab <- tapply(d$Catch_lbs, list(d$Fleet, d$Area), sum, na.rm = T)/10^6
barplot(tab, beside = T, col = c("navy", "blue", "darkturquoise", "red", "pink")
        , legend = rownames(tab),
        main = "Total dolphin catch by area and fleet",
        ylab = "total catch (millions of pounds)")
```



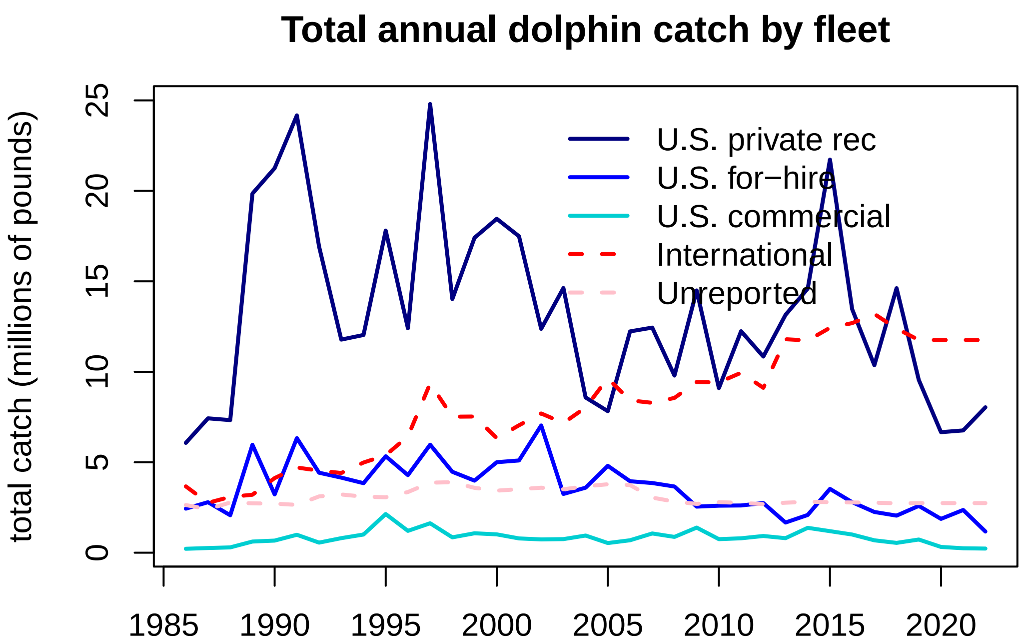
As expected, U.S. recreational fishing (private and for-hire) only takes place in the four U.S. EEZ regions (FLK, NCFL, NNC and VBM). The U.S. commercial fishery operates largely in those four U.S. EEZ regions, with small amounts of catch in the high seas of NCA, CAR and NED regions. International catches occur largely in the Caribbean, with lesser amounts in the NCA and NED high seas regions.

```
tab <- tapply(d$Catch_lbs, list(d$Quarter, d$Area), sum, na.rm = T)/10^6
tabp <- apply(tab, 2, function(x) x / sum(x, na.rm = T))

par(mar = c(2, 4, 3, 1))
barplot(tabp, beside = T, col = rainbow(4, end = 0.8),
        legend = c("winter (DJF)", "spring (MAM)", "summer (JJA)", "fall (SON)"),
        args.legend = list(x = 15, y = 0.7, bty = "n"),
        main = "Total dolphin catch by area and season",
        ylab = "proportion of catch")
```



This plot shows the seasonal movements of dolphin across the different regions, as expressed through the proportion of catch that occurs in each quarter within each area. In the NCA and CAR regions, most of the catch occurs in winter, whereas in the U.S. South Atlantic regions, most of the catch occurs in spring and summer. In the U.S. Mid-Atlantic the catch also occurs largely in spring, and also fall. In the NED region most of the catch occurs in summer.

```
tab <- tapply(d$Catch_lbs, list(d$Year, d$Fleet), sum, na.rm = T)/10^6

par(mar = c(2, 4, 3, 1))
matplot(rownames(tab), tab,
        col = c("navy", "blue", "darkturquoise", "red", "pink"),
        type = "l", lwd = 2, lty = c(rep(1, 3), 2, 2),
        main = "Total annual dolphin catch by fleet", xlab = "",
        ylab = "total catch (millions of pounds)")
legend(2002, 25, lwd = 2, lty = c(rep(1, 3), 2, 2), bty = "n",
       c("U.S. private rec", "U.S. for-hire", "U.S. commercial",
         "International", "Unreported"),
       col = c("navy", "blue", "darkturquoise", "red", "pink"))
```



This plot shows the time series of total annual dolphin catches by fleet. We can compare these figures with the plots from the previous chapters to ensure the data were processed correctly. The magnitude and variability of the catches match the expectations.

```
# output the final concatenated data file
write.csv(d, file = "data/FINAL_files/complete_catch_dataset_09052025b.csv", row.names = FA
```